Greedy Packing of Nested Rings: Superadditivity, Placement Rules, and Golden and Tribonacci Thresholds

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Abstract

We study packings of annuli ("rings") of common width into a disk, where a ring may nest inside the hole of a strictly larger one – a circular, selection-oriented relative of the Recursive Circle Packing Problem. The two natural objectives, number of placed rings and total contact area, genuinely diverge: area is a superadditive function of the radius, cardinality is not. For *superincreasing* radii (each exceeding the sum of all smaller ones) we prove that the descending greedy maximizes every positive, strictly increasing, superadditive objective, and – our main structural theorem – that the placement rule is irrelevant: any choice among feasible containers yields the lexicographically maximal feasible set, for containers of arbitrary shape and in every dimension. Both hypotheses are sharp: placement irrelevance holds unconditionally for at most three rings and fails at four, and *twin instances* rule out every placement rule that is a function of the observable state. Measuring the violation of superincreasingness by $\rho = \max_i(\sum_{j>i} r_j)/r_i$, the additive relaxation of the model has universal threshold exactly $\rho = 1$, while the geometric model resists further: we prove, with no tangency idealization, that the four-ring rigid family of counterexamples has infimum exactly the Tribonacci constant $T \approx 1.83929$ – but T is *not* the global threshold: an explicit golden family (pan $\varphi + 1$, radii $\{\varphi, 1, \varphi/2 + 2\varepsilon, \varphi/2 + \varepsilon\}$) breaks placement obliviousness at $\rho = \varphi + 3\varepsilon$, refuting the natural Tribonacci-threshold conjecture, and we conjecture that the geometric threshold is exactly the golden ratio φ, with T the exact floor of the *nested* exchange. We develop a program that proves the nested-exchange floors – exact positive-width curves with global minimum 13/7, closed profile thresholds, and blocking walls for generic containers with metallic-mean floors – and complement it with a phase diagram for the divergence of the objectives and a decoupling of hardness into a geometric layer and a combinatorial (subset-sum) layer, of which superincreasingness eliminates exactly the latter.

1 Introduction

Drop a batch of squid rings into a frying pan and a natural optimization problem appears: place as many rings as possible flat on the pan – or, if one cares about searing, maximize the total ring–pan contact area – exploiting the fact that a sufficiently small ring fits inside the hole of a

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larger one. Abstracted, this is the problem of packing annuli of common width into a disk with recursive nesting, a circular-container relative of the *Recursive Circle Packing Problem* (RCPP) introduced by Pedroso, Cunha and Tavares [1] to model the "telescoping" of tubes in shipping containers, and attacked exactly by Gleixner, Maher, Müller and Pedroso [2]. That literature is algorithmic: it packs a fixed set of rings into as few rectangular containers as possible. Here we ask structural questions instead: which objectives does the natural greedy algorithm optimize, when is the choice of *where* to put each ring irrelevant, and what conditions on the radii govern the answers.

Our starting point is a classical phenomenon in one dimension. For bin packing with *divisible* item sizes, Coffman, Garey and Johnson [3] proved that First Fit Decreasing is optimal. We are not aware of an analogue for geometric packing of disks, let alone with nesting. We prove one, with *superincreasing* radii (each radius exceeding the sum of all smaller ones) playing the role of divisibility, and we determine exactly how far the hypothesis can be relaxed – with a surprising answer: the additive relaxation of the model has threshold exactly 1, while the geometric model resists up to (conjecturally, exactly) the *golden ratio*: an explicit golden family fails at $\rho = \varphi + 3\varepsilon$, and the Tribonacci constant – the natural conjectured threshold, which our own program long targeted – turns out to be the exact floor of the *nested* exchange only.

Contributions: main theorems. (1) A *superadditivity dichotomy*: under superincreasing radii the selection greedy is optimal for every positive, strictly increasing, superadditive value function of the radius – contact area is one – while cardinality (not superadditive) admits counterexamples even under superincreasing radii (Section 3). The selection statement is best seen as a general fact about downward-closed set systems with superincreasing weights, in the lineage of greedy analyses on independence systems [13, 14] and of the superincreasing knapsack [15]; the geometric content of this paper begins with the next item. (2) A *placement-obliviousness theorem*: under superincreasing radii, *any* rule for choosing among feasible containers yields the lexicographically maximal feasible set; the proof is container-shape- and dimension-agnostic (Section 4). (3) *Sharpness*: placement obliviousness holds unconditionally for $n \leq 3$ rings and fails at $n = 4$; twin instances with an identical prefix rule out every state-based placement rule, deterministic or randomized (Section 5). (4) *Thresholds*: the additive model has universal threshold exactly $\rho = 1$; the geometric four-ring rigid subfamily has infimum exactly the Tribonacci constant T, proved with no idealization (Theorem 17, full proof in Appendix B); and an explicit *golden family* refutes the natural conjecture that the geometric threshold equals T: placement obliviousness fails at $\rho = \varphi + 3\varepsilon$ for every small ε (Theorem 19), and we conjecture the geometric threshold is exactly φ (Section 6).

Contributions: the nested-exchange program (the Tribonacci floor). (5) *The width program*: the blocked-exchange infimum with relative width $\omega > 0$ is governed by the deformed Tribonacci cubic $\alpha^3 = (1 + \omega)(\alpha^2 + \alpha + 1)$; the full lower-bound curve is in closed form and its global minimum is exactly 13/7, attained at a rational corner (Section 7); the combinatorial profile threshold is closed for every profile size, with golden plateau and exact Tribonacci crossing $\omega_T = 1/T - 1/2$. (6) *Generic containers*: an exact corona criterion for wall-tangent packings (with two corrective counterexamples and a $k = 5$ "pentagram" certificate), and a hierarchy of blocking walls – occupant holes, "blocking costs the slack", and the double pocket – yielding metallic-mean floors (silver ratio $1 + \sqrt{2}$, golden line $\varphi^2 - (\varphi/2)\omega$) that push every blocked exchange with extra occupants above T – for all widths $\omega < 0.9626$ with one extra occupant, up to 0.624 and 0.896 with two and three, and for all widths with four or more (Section 8). The remaining gaps of the program are stated explicitly there and in Open Problem 23. (7) A phase diagram for the divergence of the two objectives, and a decoupling of hardness into a geometric and a combinatorial layer, the latter with an explicit PARTITION reduction (Sections 9–10).

2 Model and preliminaries

Definition 1 (Rings and placements)**.** Fix a width $w > 0$. A *ring* of outer radius $r > 0$ is an annulus with inner (hole) radius $\max(0, r - w)$; if $r \leq w$ it is a solid disk with no hole. Let $K \subset \mathbb{R}^d$ be a compact *container* (the *pan*; our motivating case is a disk of radius R in the plane, $d = 2$, but Theorems 6 and 9 hold for arbitrary K and d, with rings read as spherical shells). A *placement* of a subset S of the rings is a forest: each placed ring has as parent either the pan or a placed ring with $r_{\text{child}} \leq r_{\text{parent}} - w$ (it sits inside the parent's hole); the children of a common parent must be packable as balls (by outer radius) with pairwise disjoint interiors inside the parent's region — K itself, or the hole ball of radius $r - w$. Feasibility is a property of the assignment (siblings may be rearranged freely inside their container).

Definition 2 (Objectives; superincreasing radii)**.** For radii $r_1 > r_2 > \cdots > r_n$, the *contact area* of a placed ring is $a(r) = \pi\big(r^2 - \max(0, r - w)^2\big)$; the two objectives are $N = |S|$ and $A = \sum_{i \in S} a(r_i)$ over feasible S. The radii are *superincreasing* if $r_i > \sum_{j>i} r_j$ for all i; the violation is measured by $\rho = \max_i \big(\sum_{j>i} r_j\big)/r_i$. The *lex-max* set L is defined greedily: scanning $i = 1, \ldots, n$, include i whenever $(L \cap \{1, \ldots, i-1\}) \cup \{i\}$ is feasible. Since feasibility is downward closed, L is the unique lexicographically maximal feasible set, and the *selection greedy* (add ring i if the resulting *set* remains feasible) computes it. Note that the selection greedy queries a *set-feasibility oracle*; it is an information bound, not an efficient algorithm – deciding sibling packability is itself hard (Section 10). The placement greedies below, by contrast, only ever attempt concrete insertions. A *descending greedy with placement rule* processes rings in decreasing radius and places each into *some* feasible container (the pan or the hole of an already placed ring), skipping it only if no container admits it.

Lemma 3 (Row lemma)**.** *Balls with radii $\rho_1, \ldots, \rho_k$ satisfying $\sum_j \rho_j \leq C$ pack inside a ball of radius C: place them tangent in a row along a diameter. Consequently, if a ball of radius m is removed from any packing, any collection of balls of total radius at most m can be inserted into the vacated ball without disturbing the rest.*

Proof. With centers on a diameter at coordinates $x_j = -C + 2\sum_{l<j} \rho_l + \rho_j$, each satisfies $|x_j| \leq C - \rho_j$. ☐

3 The superadditivity dichotomy

Lemma 4. *The contact area a is increasing and superadditive on $(0, \infty)$: $a(x) + a(y) \leq a(x+y)$.*

Proof. Monotonicity is clear. For $x, y \geq w$: $\pi w(2x - w) + \pi w(2y - w) = \pi w\big(2(x+y) - 2w\big) \leq \pi w\big(2(x+y) - w\big)$. For $x, y \leq w$ with $x + y \leq w$: $x^2 + y^2 \leq (x + y)^2$. For $x, y \leq w < x + y$: $x^2 + y^2 \leq w(x + y) \leq 2w(x + y) - w^2$, using $x^2 \leq wx$, $y^2 \leq wy$ and $x + y \geq w$. For $x \leq w \leq y$: $\pi x^2 + \pi w(2y - w) \leq \pi w(2x + 2y - w)$ since $x^2 \leq 2wx$. ☐

Lemma 5 (Lexicographic dominance)**.** *Let the radii be superincreasing and let v be any positive, increasing, superadditive function of the radius. If S, T are feasible sets and the first index (in decreasing radius order) at which they differ belongs to S, then $\sum_S v(r_i) \geq \sum_T v(r_i)$, with equality only if $T \subseteq S$.*

Proof. Let i be the first differing index and $P = S \cap \{1, \ldots, i-1\} = T \cap \{1, \ldots, i-1\}$. If $T \setminus P = \emptyset$ then $T \subseteq S$ and monotonicity of the sum under inclusion suffices. Otherwise $T \setminus P \subseteq \{i{+}1, \ldots, n\}$, so $\sum_{T \setminus P} r_j < r_i$ by superincreasingness, and by iterated superadditivity and monotonicity $\sum_{T \setminus P} v(r_j) \leq v\big(\sum_{T \setminus P} r_j\big) < v(r_i)$. Hence $\sum_T v = \sum_P v + \sum_{T \setminus P} v < \sum_P v + v(r_i) \leq \sum_S v$. □

Theorem 6 (Selection greedy; arbitrary container, any dimension)**.** *With superincreasing radii, the selection greedy computes the lex-max feasible set and therefore maximizes $\sum_{i \in S} v(r_i)$ for every positive, strictly increasing, superadditive v; in particular it maximizes the contact area A.*

Proof. Feasibility is downward closed, so the selection greedy computes L (if it could add $i \notin L$, any maximal feasible extension would lexicographically beat L). Lemma 5 shows L dominates every feasible set. □

Remark 7 (Positioning)**.** Theorem 6 is at heart a statement about arbitrary downward-closed set systems with superincreasing weights, and as such it belongs to the classical circle of ideas around greedy algorithms on independence systems [13, 14] and the superincreasing knapsack, whose lexicographic structure was analysed by Gupte [15]. The hypotheses on v are all needed: positivity rules out $v \equiv -1$ (superadditive and non-decreasing, yet maximized by the empty set), and strict monotonicity is what makes the dominance in Lemma 5 strict. The geometric novelty of this paper lies in Theorem 9, where the exchange argument must respect the geometry of nested containers.

Proposition 8 (Cardinality is not rescued)**.** *There are superincreasing instances on which every descending greedy is suboptimal for N. Example (verified): pan $R = 10$, $w = 4.8$, radii* $\{9.95, 5.0, 4.3, 0.6\}$. *Greedy places* $\{9.95, 5.0\}$ *($N = 2$, optimal area), while* $\{5.0, 4.3, 0.6\}$ *packs in a row in the pan ($N = 3$). The obstruction is structural: N corresponds to $v \equiv 1$, which is not superadditive, and Lemma 5 fails for it.*

4 Placement obliviousness under superincreasing radii

Theorem 9 (Placement rule irrelevance)**.** *Let the radii be superincreasing (weakly: $r_i \geq \sum_{j>i} r_j$ suffices) and let $K \subset \mathbb{R}^d$ be an arbitrary container. Every descending greedy, with an* arbitrary *rule for choosing among feasible containers, places exactly the lex-max feasible set. In particular best fit, worst fit and random placement are all optimal for every positive, strictly increasing, superadditive objective.*

Proof. Let F denote the greedy's placement just before processing ring i, with placed set G, and suppose $G \cup \{i\}$ is feasible via a witness placement P; it suffices to show some container of F admits i (the converse direction is immediate, and then greedy's set equals L by induction as in Theorem 6). The available containers – the pan plus one hole per ring of G – coincide in F and P, and in P the smallest ring i parents nobody.

Let m be the largest ring of G assigned different containers by F and P (if none, P restricted to G equals F and i's container in P witnesses admission). Let u be F's container for m and v

be P's. Since the greedy processes rings in decreasing order, when F placed m the occupants of u were exactly the rings larger than m that F assigns to u; by maximality of m these coincide with the rings larger than m that P assigns to u, and F certified that this set together with m packs in u. Build P' from P: (i) every ring smaller than m that P kept in u is moved into the ball of radius r_m vacated by m inside v – they fit in a row by Lemma 3, because their total radius is at most $\sum_{j:\, r_j<r_m} r_j \le r_m$, and they touch nothing else in v; (ii) m moves to u, joining exactly the larger-than-m rings whose compatibility F certified. Nesting constraints persist: u and v are the pan or holes of rings larger than m, which do not move, and every ring moved into v is smaller than r_m, which fitted in v. Thus P' is a feasible witness agreeing with F on all rings of radius $\ge r_m$: the first disagreement strictly shrinks. Iterating at most $|G|$ times yields a witness P^* agreeing with F on all of G and placing i in some container c^*; since the occupants of c^* in P^* are exactly those in F, the greedy admits i there. $\square$

Remark 10. The proof uses only (a) downward closure of packings, (b) the row lemma applied *inside the vacated ball*, and (c) the total-radius bound from superincreasingness. Nothing refers to the shape of K or the dimension: the theorem covers rectangular pans (griddles) and spherical shells nested in arbitrary containers in $\mathbb{R}^d$ verbatim. Computational corroboration of the strongest consequence: on 100 random superincreasing instances, best-fit, worst-fit and random placement produced identical, optimal outcomes without exception.

5 Sharpness: the $n = 4$ transition and twin instances

Proposition 11 ($n \le 3$, disk pan)**.** *For a disk pan and arbitrary radii, every descending greedy on at most three rings places the lex-max set.*

Proof. The only branching point is ring r_2, when both the pan ($r_1 + r_2 \le R$; two circles fit in a disk iff their radii sum to at most its radius) and the hole of r_1 ($r_2 \le r_1 - w$) admit it. Let G be the greedy's set after r_2 and suppose $G \cup \{r_3\}$ is feasible via a witness P; we check that the greedy's placement F admits r_3 in each of the four combinations of choices. If F nested r_2: should P nest it too, the containers of F and P have identical occupants and P's home for r_3 works in F; should P keep r_2 in the pan, then $r_1 + r_2 \le R$, so $r_1 + r_3 < R$ and F's pan (holding only r_1) admits r_3. If F kept r_2 in the pan: should P do likewise, the placements agree; should P nest r_2, then $r_2 \le r_1 - w$, hence $r_3 < r_2 \le r_1 - w$ and F's *empty* hole of r_1 admits r_3. In every case r_3 is admitted, and rings outside the lex-max set are never admissible, so the greedy's set is the lex-max set. $\square$

Lemma 12 (Cap confinement)**.** *Circles of radii* 10, 4.9, 4.8 *do not pack in a disk of radius* 15.

Proof. Write c_A, c_X, c_Y for the centers, measured from the pan center. Then $|c_A| \le 5$, $|c_X| \le 10.1$, $|c_Y| \le 10.2$, $|c_A - c_X| \ge 14.9$, $|c_A - c_Y| \ge 14.8$, $|c_X - c_Y| \ge 9.7$. From $|c_A| + |c_X| \ge 14.9$, $a := |c_A| \in [4.8, 5]$. Let $u = -c_A/|c_A|$. Then $c_X \cdot u = \big(|c_X - c_A|^2 - |c_X|^2 - a^2\big)/(2a) \ge (120 - a^2)/(2a) \ge 9.5$ (decreasing in a), so the transverse part of c_X has norm at most $\sqrt{10.1^2 - 9.5^2} \le 3.43$; similarly $c_Y \cdot u \ge (115 - a^2)/(2a) \ge 9.0$ with transverse norm at most $\sqrt{10.2^2 - 9^2} = 4.8$. Hence $|c_X - c_Y|^2 \le 10.1^2 + 10.2^2 - 2(9.5 \cdot 9.0 - 3.43 \cdot 4.8) \le 67.98$, i.e. $|c_X - c_Y| \le 8.25 < 9.7$, a contradiction. $\square$

Theorem 13 (Failure at $n = 4$)**.** *Placement obliviousness fails for four rings: with* $R = 15$*,* $w = 0.3$ *and radii* $\{10, 5, 4.9, 4.8\}$ *(all four form the lex-max set, witnessed by the pan pair*

$\{10, 5\}$ *at exact tangency and the hole pair* $\{4.9, 4.8\}$ *filling the hole* 9.7 *exactly), best fit nests the* 5*, forces the* 4.9 *into the pan, and blocks the* 4.8 *everywhere by Lemma 12 and the exact two-circle conditions; worst fit places all four.*

Theorem 14 (Twin instances; no state-based rule)**.** *Let* $R = 15$*,* $r_1 = 10$*,* $r_2 = 5$*,* $w = 0.505$ *(hole* 9.495*), and*

$$I_1 = \{10,\ 5,\ 4.99,\ 4.50\}, \qquad I_2 = \{10,\ 5,\ 4.76,\ 4.74\}.$$

In I_1 *the correct placement of the* 5 *is the pan (best fit fails, worst fit succeeds); in* I_2 *it is the hole (worst fit fails, best fit succeeds). At the decisive step the observable state – containers, capacities, occupants, incoming ring,* R *and* w *– is identical in* I_1 *and* I_2*. Consequently no deterministic placement rule that is a function of the state attains the lex-max set on all instances, and for every randomized rule some instance is failed with probability at least* 1/2*.*

Proof ingredients. Feasibilities are two-circle-exact except for two triples, both settled rigorously: $\{10, 4.99, 4.50\}$ does not pack in the disk of radius 15 (cap confinement as in Lemma 12 yields $|c_X - c_Y| \le 8.80 < 9.49$), and the pan pair $\{10, 5\}$ is diametrically rigid, whence no third circle of radius ≥ 4.7 fits (confinement again: for $z = 4.76$, $|c_z - c_B| \le \sqrt{204.86 - 20 \cdot 8.8} \le 5.37 < 9.76$). Packability of $\{10, 4.76, 4.74\}$ in radius 15 is exhibited by an explicit boundary configuration (angles $180°$, $30.8°$, $-32.0°$; all pairwise constraints verified). The key phenomenon enabling the twins is that triple-packability is *not monotone in the sum*: the unbalanced pair $\{4.99, 4.50\}$ (sum 9.49) fails beside the 10 while the balanced pair $\{4.76, 4.74\}$ (sum 9.50) succeeds. $\square$

Remark 15. The obstruction is informational, not existential: following any witness of the lex-max set is itself a legal greedy execution (a ring outside the lex-max set is never admissible), so some execution always succeeds; what cannot exist is a state-based rule that finds it. For rules with access to the full input the question becomes one of computational complexity and remains open (Section 12).

6 Thresholds: the Tribonacci floor and the golden counterexample

Theorem 16 (Additive model: threshold exactly 1)**.** *Replace sibling feasibility by its additive surrogate (a set of siblings fits iff their radii sum to at most the container capacity; exact for at most two circles and sufficient in general by Lemma 3). Then the* universal *threshold for placement obliviousness is exactly* 1*: obliviousness holds for* all *instances with* $\rho \le 1$ *(the proof of Theorem 9 applies verbatim), while for every* $\varepsilon > 0$ some *instance with* $\rho \le 1 + \varepsilon$ *fails – individual instances with larger* ρ *may of course still succeed. Indeed, with* r_1 *fixed,* $s = R - r_1 \in \left(\frac{2}{3}(r_1 - w),\, r_1 - w\right]$*,* $w > s/2$*, take* $r_2 = s$ *and a pair* $r_3 \approx r_4 \approx s/2$ *with* $r_3 + r_4 = s + \varepsilon$*: the witness fills the pan with* $\{r_1, r_2\}$ *and the hole with the pair, while best fit nests* r_2*, sends* r_3 *to the pan, and blocks* r_4 *additively (*$r_1 + r_3 + r_4 > R$*;* $r_2 + r_4 > r_1 - w$*;* $r_4 > r_2 - w$*). All ratios tend to* 1*; verified instances reach* $\rho = 1.03$ *and* 1.02*.*

The four-ring counterexample family carries a genuine floor at the Tribonacci constant. An earlier version of this statement relied on a tangency idealization ($r_3 \to r_2$, $w \to 0$); the following theorem removes it entirely: the rigid configuration is not assumed but *emerges* as the extremal case of a concavity argument.

Theorem 17 (Rigid-subfamily Tribonacci floor, no idealization)**.** *Normalize $r_1 = 1$ and write $t = r_2$, $u = r_3$, $v = r_4$, $\omega = w > 0$. Let $\mathcal{F}$ be the family of four-ring instances with* (F1) *diametral pan $R = 1 + t$,* (F2) *$u + v \le 1 - \omega$ (the witness pair fits the hole of r_1), and* (F3) *the triple $\{1, u, v\}$ does not pack in the disk of radius $1 + t$. Then every instance of $\mathcal{F}$ satisfies $\rho > T$, strictly, for every width $\omega > 0$ and all strict radii $v < u < t < 1$; moreover $\inf_{\mathcal{F}} \rho = T$ and the infimum is not attained. The family is nonempty only for $t < t^* = 1/T$, and it contains the $n = 4$ counterexample and the twin I_1 of Theorem 14.*

Proof sketch (complete proof in Appendix B). For two circles of radii a, b internally tangent to the wall of a disk of radius R, the minimal angular separation $\theta(a, b)$ obeys the half-angle identity $\sin^2\bigl(\theta/2\bigr) = f(a)f(b)$ with $f(x) = x/(R - x)$. Placing the triple $\{1, u, v\}$ wall-tangent with the unit ring in the middle packs it whenever $\theta(1, u) + \theta(1, v) + \theta(u, v) \le 2\pi$ (the two non-adjacent arguments are separated by monotonicity of θ), and this angular condition reduces *algebraically* to

$$\psi(u) + \psi(v) \;\ge\; \tau, \qquad \psi(x) = \sqrt{\tfrac{t-x}{x}}, \qquad \tau = \tfrac{t}{\sqrt{1+t}},$$

(an algebraic *sufficient* condition for packing, which is all the contrapositive needs: (F3) forces $\psi(u) + \psi(v) < \tau$). In the coordinates $\alpha = \psi(u)$, $\beta = \psi(v)$ the sum $u + v = U(\alpha) + U(\beta)$ with $U(z) = t/(1 + z^2)$, and U is concave on $[0, \tau]$ exactly when $\tau \le 1/\sqrt{3}$, i.e. $t \le (1 + \sqrt{13})/6 = 0.7676\ldots$, which covers the relevant range with room to spare; hence the infimum of $u + v$ over the blocked (open) region sits at the *corner* of its closure, $u = t$, $v = b(t)$, where $b(t) = t(1 + t)/(1 + t + t^2)$ is the Descartes pocket of the rigid pair. This gives the two pressures $t + b(t) < u + v \le 1 - \omega < 1$ (the first one strict, whence the strict floor and the non-attainment), whose compatibility is precisely the cubic $t^3 + t^2 + t < 1$, and then $\rho \ge (u+v)/t > 1+(1+t)/(1+t+t^2) =: L(t)$ with L strictly decreasing and $L(t^*) = T$. Exactness of the pocket at the rigid pair (needed only for the direction $\inf \le T$) is a rigidity computation: the constraint left-hand side factors exactly as $4(t^2 + t + 1)\bigl(b(t) - v\bigr)$; and the approximating family lives genuinely inside $\mathcal{F}$ by a closure-and-monotonicity lemma for the packing region (compactness of witnesses), which shows the blocked interval in u is half-open with an attained endpoint. $\square$

Remark 18 (Slack pans)**.** The hypothesis $R = r_1 + r_2$ can be relaxed to $R \ge r_1 + r_2$ at no cost: non-packability is antitone in the container (a packing in a smaller disk is a packing in a larger one after translation), so (F3) at radius R implies (F3) at radius $r_1 + r_2$ and Theorem 17 applies verbatim; the infimum remains T. Slack in the pan never helps the blocker — what it changes is the set of possible *occupants*, the subject of Section 8.

The natural conjecture – which we ourselves held until late in this project – is that the geometric threshold is exactly T. It is *false*. The witness capacity that lifts the floor from φ to T in Theorem 17 exists only when the greedy's container for the pivotal ring is a *hole*; when it is the pan itself, the floor drops to the first rung of the ladder, the Descartes pocket, and the golden ratio appears:

Theorem 19 (The golden family: the threshold is at most φ)**.** *For every $\omega \in (1-\varphi/2,\ \varphi-1) = (0.1910, 0.6180)$ and every sufficiently small $\varepsilon > 0$, the instance*

$$R = \varphi + 1, \qquad w = \omega, \qquad r = \bigl\{\varphi,\ 1,\ \tfrac{\varphi}{2} + 2\varepsilon,\ \tfrac{\varphi}{2} + \varepsilon\bigr\}$$

has $\rho = \varphi + 3\varepsilon < T$, its lex-max set is all four rings, and the descending greedy under worst fit places only three: placement obliviousness fails at $\rho = \varphi + 3\varepsilon$.

Proof. Write $s_1 = \frac{\varphi}{2} + 2\varepsilon$, $s_2 = \frac{\varphi}{2} + \varepsilon$. Every feasibility below is exact. *The value of* ρ: the tails are $(1 + s_1 + s_2)/\varphi = \varphi + 3\varepsilon/\varphi$ (by the golden identity $(1 + \varphi)/\varphi = \varphi$), $s_1 + s_2 = \varphi + 3\varepsilon$ (dominant) and $s_2/s_1 < 1$, and $\varphi + 3\varepsilon < T$ for $\varepsilon < (T - \varphi)/3$. *The witness*: place 1 in the hole of φ (capacity $\varphi - \omega \geq 1 \iff \omega \leq \varphi - 1$) and $\{\varphi, s_1, s_2\}$ as a corona in the pan: their angular sum is $5.004 < 2\pi$ with slack 1.28 as $\varepsilon \to 0$ (explicit coordinates in the repository), so all four rings are placeable. *The greedy under worst fit*: $\varphi \to$ pan; the ring 1 fits both the pan ($\varphi + 1 \leq R$, exact tangency, as in Theorem 13) and the hole of φ, and worst fit – the container of largest capacity, $R = \varphi + 1 > \varphi - \omega$ – chooses the *pan*. Now $\{\varphi, 1\}$ fills the pan diametrally: the pair is rigid and, by the exact necessity of Proposition 30 (rescaled), any third circle has radius at most $b_2(\varphi, 1) = \varphi/2$ – the golden pocket identity. Hence neither s_1 nor s_2 fits the pan; the hole of φ admits one of them but not both ($s_1 + s_2 = \varphi + 3\varepsilon > \varphi - \omega$, two-circle exact); the hole of m admits neither ($s_2 > 1 - \omega \iff \omega > 1 - \frac{\varphi}{2} - \varepsilon$); and s_2 cannot nest in s_1. An exhaustive placement tree confirms the count: three rings with 1 in the pan, four with 1 in the hole (`code/aureo.py`). □

The instance is golden four times over: the occupant is φ (the unique positive fixed point of $2b(A)\,A = 1 + 2b(A)$), the pocket is $b_2(\varphi, 1) = \varphi/2$, the tail of φ is $(1 + \varphi)/\varphi = \varphi$, and $\rho \to \varphi$. The family is not degenerate: the ω-window has width 0.427, ε ranges over an interval, and the diametral tangency survives pan slack up to $\delta^* = 0.0248$ (numerically, via the angular criterion). It escaped the earlier structured searches because it lives at a codimension-one corner (pan tangency) that random radii never visit.

Conjecture 20 (Golden threshold)**.** *Placement obliviousness holds for every instance with $\rho < \varphi$; together with Theorem 19 (which realizes every $\rho \in (\varphi, T)$) and the failures above T, this would make the geometric threshold exactly the golden ratio: $\varphi - 1$ measures the protection that disk geometry grants the greedy over additive bookkeeping, and T is the exact floor of the* nested *exchange (Sections 7–8), attained by the rigid family of Theorem 17. Evidence: the pan-exchange φ-floor is proved in the repository for pair profiles except a sliver (Open Problem 23(a)) – one occupant at every width via the golden fixed point, the evacuation branch at every width via $\Psi_B(1) = \varphi$, and the strict-leaf ladder with $\Psi_1(\frac{1}{2}) = \Psi_2(\varphi/2) = \varphi$ – and every proved floor of the nested exchange exceeds $T > \varphi$ with margin.*

7 The width program: profile thresholds and the 13/7 corner

Towards Conjecture 20, the proof of Theorem 9 localizes all difficulty in a single *exchange step*: m is the largest ring that the greedy F and a witness P place in different containers $u = c_F(m)$, $v = c_P(m)$, and the set S of rings smaller than m that P keeps in u must be reinserted using the resources freed by moving m to u: the vacated disk D_m (radius r_m), the hole H_m (capacity $r_m - w$, travelling with m), nesting, and the geometry of v. Normalize $r_m = 1$ and $\omega = w/r_m$. Everything ρ imposes on S is the pair of inequalities $\sum S \leq \rho$ and $\sum_{l>j} x_l \leq \rho\, x_j$; a lower bound for ρ over all *blocked* exchanges is therefore a lower bound for any failure of placement obliviousness. Two regimes must be distinguished. When u is the *hole* of a ring (the *nested* exchange), the witness placement supplies the capacity wall $\sum S \leq \mathrm{cap}(u)$, and this section and the next prove floors that all exceed T: the Tribonacci constant is the exact threshold of the nested exchange. When u is the pan, that wall is absent, the floor drops to the Descartes pocket, and the golden family of Theorem 19 realizes it: the *pan* exchange governs the global threshold of Conjecture 20. Complete proofs for every statement of this section are given in Appendix C.

The combinatorial layer is closed. For two-ring profiles the blocked threshold is exactly $\rho_2^*(\omega) = \max\bigl(1,\, 2(1-\omega)\bigr)$; in particular no two-ring blocking is compatible with $\rho < T$ for $\omega \le 1 - T/2 = 0.0804$. For three rings the threshold is closed,

$$\rho_3^*(\omega) = \max\Bigl(1,\ \min\bigl(2(1-\omega),\ \max\bigl(\varphi,\ \tfrac{2}{1+2\omega}\bigr)\bigr)\Bigr),$$

with a purely additive proof and a *golden plateau* $\rho_3^* \equiv \varphi$ on $\omega \in \bigl[\frac{\sqrt{5}-2}{2},\, 1-\frac{\varphi}{2}\bigr]$; and a general additive tree argument shows $\rho_k^* = \rho_3^*$ for *every* $k \ge 3$. Consequently the combinatorial threshold crosses T exactly at

$$\omega_T \;=\; \tfrac{1}{T} - \tfrac{1}{2} \;=\; T^2 - T - \tfrac{3}{2} \;=\; 0.043689\cdots:$$

below ω_T the exchange step never blocks with $\rho < T$, with no geometry at all (Propositions 33–36, Theorem 35, Corollary 37).

The canonical template with positive width. For $\omega > \omega_T$ geometry enters through the canonical template (v a diametral pan with one big neighbour α; u the hole of α, capacity $\alpha-\omega$; $S = \{\sigma_1 \ge \sigma_2\}$ placed there by the witness). Blocking must defeat four resources, giving the walls $\sigma_2 > 1-\omega$ (H_m), $\sigma_2 > \alpha-\omega-1$ (σ_2 next to m inside u), $\sigma_1+\sigma_2 \le \alpha-\omega$ (the witness), and non-packability of the triple $\{\alpha, \sigma_1, \sigma_2\}$ in the pan. Two structural facts organize the optimum. First, on the blocking frontier of the triple one has the closed form $t(\sigma_1) + t(\sigma_2) = t(b(\alpha))$ with $t(s) = \sqrt{(1-s)/s}$ (the Descartes pocket read in the t-coordinate), and the frontier slope satisfies the α-*independent* identity

$$\kappa \;=\; -\frac{\partial \sigma_2}{\partial \sigma_1} \;=\; \sqrt{\frac{g(\sigma_2)}{g(\sigma_1)}}, \qquad g(s) = s^3(1-s), \qquad \kappa \ge 1 \text{ for all } \alpha > 1,$$

so the cheapest blocked pair sits at $\sigma_1 \to 1$, $\sigma_2 \to b(\alpha)$; moreover $1 + b(\alpha) \ge \alpha$ iff $\alpha \le T$, so no blocking exists at all for $\alpha \le T$. Second, the witness capacity deforms the cubic: with T_c the positive root of $\alpha^3 = c(\alpha^2+\alpha+1)$, the witness branch has infimum $\Phi(\omega) = T_{1+\omega} - \omega$, strictly increasing and concave with $\Phi'(\omega) = (2\alpha+1)/\bigl(\alpha^2(\alpha^2+2\alpha+3)\bigr)$ at $\alpha = T_{1+\omega}$. The full curve of the blocked infimum $T_{\mathrm{can}}(\omega)$ is closed *as a proved lower bound* — the matching upper bound is exact on the witness branch and at the corner below, and inherits the usual angular-criterion caveat on the other two branches: $2(1-\omega)$ on $(0, \omega_1]$, a sextic-algebraic middle branch on $[\omega_1, 1/7]$, and Φ on $[1/7, 0.30]$, where $\omega_1 = 0.0413570\ldots$ is the root of $4\omega^3 - 20\omega^2 + 25\omega - 1$ in $(1/25, 1/14)$ (Lemma 38, Theorem 39, Corollary 40, Propositions 41 and 43).

Theorem 21 (The 13/7 corner)**.** *For all $\omega > 0$,*

$$T_{\mathrm{can}}(\omega) \;\ge\; \tfrac{13}{7} \;=\; T + 0.017856\ldots,$$

with equality exactly in the limit at the rational corner $(\omega, \alpha, \sigma_2) = \bigl(\frac{1}{7},\, 2,\, \frac{6}{7}\bigr)$, where all four walls saturate simultaneously (note $T_{8/7} = 2$ because $2^3 = 8 = \frac{8}{7}\cdot 7$). The bound is unconditional – in particular it does not depend on the exactness of any angular feasibility criterion – and the approximating family is genuine. Thus positive width only raises *the Tribonacci floor, uniformly.*

The fine structure is amusing: the curve is *not* monotone on $(0, 1/7]$ – ω_1 is a local minimum, followed by a bump of height $1.1 \cdot 10^{-4}$ at $\omega_{\mathrm{peak}} \approx 0.0445$ (a root of an explicit degree-8 polynomial) before descending to the corner.

8 Generic containers: coronas, walls, and metallic floors

The canonical template assumes v contains only the big neighbour and m. The remaining gap towards Conjecture 20 is genericity of the containers, and it is attacked by a toolkit of independent walls, each adversarially verified. Throughout, v is a pan of radius R with occupants $\{\alpha\} \cup O \cup \{m\}$, $O = \{o_1 \geq \cdots \geq o_j\}$, $o_i \geq m = 1$, u is the hole of α, $S = \{\sigma_1 \geq \sigma_2\}$, and holes may be occupied arbitrarily, at any nesting depth. Complete proofs for every statement of this section are given in Appendix D.

The universal trio frontier. For a trio $\{A, x, y\}$ wall-tangent in a disk of *arbitrary* radius R (interior domain, and under the necessary hypothesis $A \geq \min(x, y)$), the angular blocking condition is *linear* in the coordinates $T_c(x) = \sqrt{(c-x)/x}$, $c = R - A$: θ-infeasibility $\iff$ $T_c(x) + T_c(y) \leq \tau_R = c/\sqrt{AR}$. The pocket generalizes to $b_R(A) = ARc/(AR + c^2)$, increasing in R, and the frontier slope identity $\kappa = \sqrt{g_c(y)/g_c(x)}$, $g_c(s) = s^3(c-s)$, is uniform in R. Without the hypothesis $A \geq \min(x, y)$ the equivalence is false (an explicit counterexample pins the failure region exactly), a fact found by adversarial verification (Lemma 45).

An exact corona criterion. Call a *corona* a packing of k circles all wall-tangent. For a fixed cyclic order the corona constraints form a linear feasibility system in the angular gaps, so coronas are decidable exactly; every subset of a corona yields a necessary *certificate* (its induced cyclic sum of pairwise angles is at most 2π). For $k = 4$ the certificates are exactly sufficient – for arbitrary symmetric separations in $(0, \pi]$, by a negative-cycle argument: a violating cycle with U ascents needs more than $2U$ edges, so on four nodes only $U = 1$ cycles can violate, and those are exactly the subset certificates – and the global criterion collapses to *two* inequalities: the top trio, plus the total of the *zigzag* order (a_1, a_3, a_2, a_4), which minimizes the cyclic total by a convexity/majorization argument ($\theta = g(\log f(a) + \log f(b))$ with g convex). Two natural guesses are refuted by explicit counterexamples: the consecutive-arc condition alone is *not* sufficient for $k \geq 4$, and the sorted order is *not* optimal (the zigzag is: large circles must not be angular neighbours). For $k = 5$ sufficiency requires exactly one additional certificate, the *pentagram* (the five-diagonal star $\{5/2\}$, with combinatorially sharp example $\theta = \pi$ on the diagonals), conjecturally redundant for separations of geometric origin (Lemmas 46–47, Theorems 48–50).

Occupant walls and metallic floors. Each extra occupant's hole is itself a reinsertion resource, and blocking it is *taxed*. The basic count: with free holes, blocking forces every extra occupant to within one width of m ($o_i < \sigma_2 + \omega$), and the largest occupant's tail gives $\rho > (j+2)/(1+\omega)$ – each occupant pays $1/(1+\omega)$ – whence, with a refined branch $\rho > 4/(1+2\omega)$ for $\omega \geq \frac{1}{2}$, $\rho > T$ for $\omega < 2/T - \frac{1}{2} = 0.5874$ when $j = 1$ and for *all* ω when $j \geq 2$; adding occupants is strictly suboptimal for the blocker, since the canonical curve satisfies $\Phi(\omega) < 2$ identically (the polynomial identity $(2+\omega)^3 - (1+\omega)\big((2+\omega)^2 + (2+\omega) + 1\big) = 1$). For occupied holes the tax is quantified by the *opposite-disk lemma*: in a disk of capacity c with largest piece σ, if the remaining mass is at most $c - \sigma$ the whole family packs (the wall-tangent σ leaves a free disk of radius exactly $c - \sigma$ opposite, filled in a row); hence blocking a hole costs at least its slack, $y < \sigma_2 + \omega + X_y$ with X_y the hole content, for every *extra* occupant o_i and every ring of radius $\geq r_m$ nested inside one (α and m themselves are excluded: α's hole is u, governed by its own walls, and excluding m is essential). Following the *minimal* such ring and balancing two tails yields, for arbitrary hole occupancy at any depth,

$$\rho \;>\; \Psi(\omega) \;=\; (1-\omega) + \sqrt{(1-\omega)^2 + 1},$$

the metallic mean of $2(1-\omega)$: $\Psi(0) = 1+\sqrt{2}$ is the silver ratio, $\Psi(1/4) = 2$ exactly, and $\Psi > T$ iff $\omega < (T-1)^2/2 = 0.3522$. Allowing m itself to carry children splits by an evacuation dichotomy into the same bound and a second metallic mean, the root of $u^2-(2-\omega)u-1$, which dominates automatically; its own crossing is at $(T-1)^2$ exactly – twice the previous one, and the certifying identity *is* the Tribonacci polynomial: $(T-1)^2\,T-(2T-T^2+1) = T^3-T^2-T-1$ (Lemmas 51 and 54, Proposition 52, Theorems 56 and 57).

The double-pocket wall and the golden line. The geometric wall needs no angular criterion at all: blocking implies the four rings $\{\alpha, o_1, \sigma_1, \sigma_2\}$ do not pack in the pan, hence not in the disk of radius $\bar{R} = \alpha + o_1$ (containment), where the pair $\{\alpha, o_1\}$ is diametrically rigid and leaves exactly *two* Descartes pockets $b_2(\alpha, o_1) = \alpha o_1(\alpha+o_1)/(\alpha^2+\alpha o_1+o_1^2)$, one on each side (their mirror centers are $4b_2$ apart: the exact identity $y_0 = 2b_2$). Thus blocking forces $\sigma_1 > b_2(\alpha, o_1)$. Feeding this wall into the program of the previous walls produces a *golden* optimum: the binding corner is $\alpha = 2$, $o_1 = \sqrt{5}-1$ – the pair with unit pocket, $b_2(2, \sqrt{5}-1) = 1$ exactly, self-dual in the sense $A_{\max}(\sqrt{5}-1) = 2$, $A_{\max}(2) = \sqrt{5}-1$ – and, for one extra occupant and *any* hole occupancy, both branches of the analysis meet the same line: $\rho > T$ unconditionally for all $\omega < \omega_A$, via

$$\rho \;>\; \varphi^2 - \tfrac{\varphi}{2}\,\omega, \qquad \omega_A \;=\; 2-2(T-1)(\varphi-1) \;=\; 0.962585\ldots$$

(the golden line itself is proved in all but one corner configuration – branch B with a nested node inside the occupant and $\alpha < o_1$, possible only for $\omega < \frac{1}{2}$ – where the proved bound is $\geq 2 > T$, via $\Psi_B(\frac{1}{2}) = 2$ exactly). The proof combines two exact facts – b_2 is concave in α ($\partial^2 b_2/\partial\alpha^2 = -6\alpha o^3(\alpha+o)/D^3$) and $D^2 - \alpha o^2(o+2\alpha) = (\alpha+o)(\alpha^3+\alpha^2 o+o^3) > 0$ – with one-variable boundary certificates anchored at the identity $f_1 \equiv$ curve at $o_1 = \sqrt{5}-1$. For $j \geq 2$ occupants the ladder $\Psi_j = (1-\omega)+\sqrt{(1-\omega)^2+j}$ (crossing T at $1-(T^2-j)/(2T)$: 0.624 for $j = 2$, 0.896 for $j = 3$, all ω for $j \geq 4$) covers the corresponding ranges unconditionally, for arbitrary nested occupancy of the holes: a *leaf argument* closes the case where occupants' holes contain further rings of radius $\geq r_m$. Every occupant subtree contains a leaf (a node none of whose hole contents reach r_m); taking L the largest of the j disjoint leaves, its tail collects the other $j-1$ leaves together with m, σ_1, σ_2 and all small mass W, while the blocking wall at a leaf reads $L < \sigma_2 + \omega + W$; optimizing the resulting two-branch program returns exactly Ψ_j, with the crossing at $\sigma_2 = 1-\omega$ – the same metallic-mean mechanism as Ψ. Small rings anywhere compatible with each theorem's template – and with S still a pair – are free for all the combinatorial walls, whose unblocking placements are local to D_m, H_m, the holes, and the slot next to m inside u (Lemma 59, Theorem 60, Proposition 61, Corollary 62).

Three-piece profiles: the zigzag pays. Larger reinsertion profiles are *cheaper* for the adversary in pure nest accounting ($\rho_3^* \leq \rho_2^*$), but in the canonical template the geometry reverses the sign: blocking with $S = \{\sigma_1 \geq \sigma_2 \geq \sigma_3\}$ and arbitrary occupancy of H_m forces $\rho > \max\bigl(\Phi(\omega), \frac{13}{7}\bigr)$ for *all* ω, with no exceptional branch. If $\sigma_3 \leq \sigma_1 - \omega$ the dust rides inside σ_1 and the pair program is inherited wall by wall (the 13/7 corner bound applies); if nothing nests and the trio $\{\alpha, \sigma_1, \sigma_2\}$ has no corona, the witness chain gives $\sigma_1+\sigma_2 \geq 1+b(\alpha) \geq \Phi(\omega)$; and if the trio *does* pack, Lemma U_4 read backwards forces the *zigzag* certificate to fail, and the zigzag implies exactly $\sigma_2 > b(\alpha)$ – the Descartes pocket by a third route, via $\sin^2 A + \sin^2 B > 1$. Crossing that wall with the witness capacity lands on the *golden deformation* γ_ω, the root of $\alpha^3 = \alpha^2 + \alpha + \omega(\alpha^2 + \alpha + 1)$ with $\gamma_0 = \varphi$, and the tail bounds meet at another rational corner: $\gamma_{2/7} = 2$ exactly $(8 = 4 + 2 + 2)$ with value $\frac{17}{7}$ – the sister of the 13/7 corner (Theorem 63, Proposition 64).

Status. Together with Theorem 21, every blocked *nested* exchange in the templates above satisfies $\rho > T$: for all ω in the canonical template – with S a pair or a triple – for $\omega < 0.9626$ with one extra occupant under arbitrary hole occupancy, for $\omega < 0.624$ and $\omega < 0.896$ with two and three, and for all ω with four or more. All these floors exceed $T > \varphi$, so for Conjecture 20 the decisive open front is the *pan* exchange (Open Problem 23); the remaining nested gaps – the width tips ($j = 1$, $\omega \geq 0.9626$; $j = 2$, $\omega \geq 0.624$; $j = 3$, $\omega \geq 0.896$), a quantitative "gap lemma" (small rings against the geometric wall; tiny σ_2), and profiles of four or more pieces beyond the reduction branch (and $k \geq 3$ with extra occupants beyond it) – matter only for pinning the nested threshold at exactly T. Each statement in this section is backed by a verification script and an adversarial verification report in the repository.

9 The two objectives diverge: a phase diagram

Contact area behaves like $2\pi w \sum r_i$ minus a per-ring penalty πw^2; cardinality counts rings. With uniform width the hole of a large ring is large, so nesting usually reconciles the two objectives, and the divergence survives only marginally: small rings that fit k times in the pan but $k - 1$ times in the large ring's hole. The minimal instance is $R = 10$, $w = 1$, radii $\{9.0, 4.2, 4.2, 4.2\}$: the area optimum is the 9.0 with one nested 4.2 ($N = 2$, $A \approx 76.7$), the cardinality optimum the three 4.2's ($N = 3$, $A \approx 69.7$). Figure 1 maps the divergence region for the family "one large ring b plus unlimited equal small rings ρ": a staircase governed by the proved optimal thresholds for n equal circles in a disk [8, 9, 10, 11, 12]; its upper edge is exactly the three-circle threshold $0.4641R$.

10 Hardness decouples into two layers

If all radii lie within a band of width less than w, no nesting is possible and the decision problem "do all rings fit?" is exactly circle packing of the outer radii into the pan.

The geometric layer. Since Theorems 6 and 9 hold for arbitrary containers, our model contains, as the special case of a *square* pan with a no-nesting band, the problem of deciding whether given circles fit into a square, which is NP-hard by reduction from 3-Partition [4]; hence the geometric layer of the model is NP-hard. Membership in NP is itself delicate, because tight packings may require coordinates without short descriptions – a phenomenon made precise by the $\exists\mathbb{R}$-completeness framework for two-dimensional packing [6]. For the disk pan specifically, we are not aware of a published NP-hardness proof (the related-work discussions of [5] and its companions attribute hardness to the square container only), and we leave its precise status as part of Open Problem 25. Integer-programming formulations for circle packing in rectangular containers are studied in [16].

The combinatorial layer. Independently of the geometry, the *additive surrogate* (Theorem 16) is weakly NP-hard, and here the reduction can be made exact despite the per-ring penalty in the area.

Proposition 22 (The additive layer is weakly NP-hard)**.** *Deciding whether an instance of the additive surrogate admits a feasible set of contact area at least a given threshold is weakly NP-hard, already for instances with a no-nesting radius band.*

Proof. Reduce from Equal-Cardinality Partition: given positive integers $a_1, \ldots, a_n$, n even, with $\sum_i a_i = 2B$, decide whether some S with $|S| = n/2$ has $\sum_S a_i = B$. (This variant is NP-hard by the standard padding: scaling a Partition instance by $2n$ and adjoining n unit items forces, modulo $2n$, every solution to use exactly n of the $2n$ items.) Fix any width $w > 0$, set $M = 3B$, pick $\delta \in \left(\frac{w}{3B}, \frac{w}{2B}\right)$, and take radii $r_i = \delta(a_i + M)$ with pan capacity $R^+ = \delta\left(\frac{n}{2}M + B\right)$. All radii lie in $[3B\delta, 5B\delta]$, so $r_i > w$ (each is a genuine ring, $a(r) = 2\pi wr - \pi w^2$) and $r_{\max} - r_{\min} = 2B\delta < w$ (no nesting): feasibility is exactly $\sum_S r_i \le R^+$. Write $s = |S|$ and $t = \sum_S a_i \le 2B$; the area is $A(S) = 2\pi w\delta(sM + t) - \pi w^2 s$, and we ask whether $A(S) \ge \Theta := 2\pi w\delta\left(\frac{n}{2}M + B\right) - \pi w^2 \frac{n}{2}$. If a balanced partition exists, taking S with $s = n/2$, $t = B$ attains Θ exactly. Conversely: $s > n/2$ is infeasible, since it would force $(s - \frac{n}{2})M \le B - t \le B < M$. For $s = \frac{n}{2} - q$ with $q \ge 1$, the requirement $A(S) \ge \Theta$ rearranges to $wq \ge 2\delta(qM + B - t)$, but

$$2\delta(qM + B - t) \;\ge\; 2\delta(qM - B) \;\ge\; 2\delta q(M - B) \;=\; 4\delta qB \;>\; wq$$

using $t \le 2B$, $q \ge 1$, $M = 3B$ and $\delta > \frac{w}{4B}$ (implied by $\delta > \frac{w}{3B}$): impossible. Finally $s = n/2$ forces $t \ge B$ from $A \ge \Theta$ and $t \le B$ from feasibility, i.e. $t = B$: a balanced partition. □

Superincreasing radii eliminate exactly this combinatorial layer – precisely as in the superincreasing knapsack [15] – leaving only the geometric one: by Theorem 6, the area optimum then reduces to n calls to a sibling-packing feasibility oracle. Useful blackboxes for the remaining geometry include the critical-density theorem for disks in a disk (any family of total area at most half the container's packs, and 1/2 is tight) [5] and Split Packing [7].

11 Generalizations

The model invites several generalizations, all of which we record here both as context for the theorems (two of them hold with no extra work) and as a research program.

(a) Container shape and dimension. Theorems 6 and 9 are stated and proved for arbitrary containers in $\mathbb{R}^d$: rectangular griddles, and tubes/spherical shells in three dimensions – the original RCPP setting – are covered verbatim, because the exchange argument acts inside the vacated ball of the moved ring only. The sharpness results (Section 5) are disk-specific; their analogues for squares and in $\mathbb{R}^3$ are open.

(b) Variable width. With per-ring widths w_i, a large ring may have a small hole, which destroys the "nesting rescue" and should widen the divergence band of Section 9 dramatically; the superadditivity lemma must be re-examined since a depends on both r and w.

(c) Flexible rings. Real squid rings bend: a ring partially resting on another still touches the pan outside the overlap, up to a lifted ramp of width δ around each overlap. The idealization $\delta = 0$ (contact area equals the annulus minus the overlapped region) defines a continuous

relaxation in which partial placements trade contact for cardinality; $\delta > 0$ penalizes overlaps by a dead band proportional to the overlap perimeter.

(d) Infinite inventories. Unlimited supplies (already used in the phase diagram) suggest degenerate limits: maximal nested chains r, $r-w$, $r-2w, \ldots$, density questions, and Apollonian-type limit configurations.

(e) Objectives. Theorem 6 covers every positive, strictly increasing, superadditive value; weighted counts and concave values fall outside and, by Proposition 8, genuinely so.

12 Open problems

Open Problem 23. Prove Conjecture 20. Two halves. (a) *The pan exchange has floor φ*: largely proved in the repository for pair profiles (adversarially verified): blocking the pan exchange forces $\rho > \varphi$ for one occupant at every width (both reinserted rings exceed the pocket $b(o_1)$ by containment and rigidity, and the two tails meet at the golden fixed point $2b(A)A = 1 + 2b(A)$, whose crossing factors as $(A^2 - A - 1)(2A + 1)$); for the evacuation branch at every width (a *strict leaf* yields the Ψ_B program and $\Psi_B(1) = \varphi$ exactly); and for the remaining branch through the strict-leaf ladder ($\Psi_1(\frac{1}{2}) = \varphi$, $\Psi_2(\varphi/2) = \varphi$, $\Psi_3(1) = \sqrt{3} > \varphi$), with a U_4-based wall ($o_2 < 1+1/o_1$) trimming what is left. Open: a sliver (two or three occupants, large width, $o_1 > 3/\varphi$; numerical evidence $\geq \varphi + 0.41$), profiles of three or more reinserted pieces, and extra small rings. (b) *Assemble the universal reinsertion lemma* for the nested exchange from the proved floors ($\geq T > \varphi$, so far above the golden threshold that the remaining nested gaps – the width tips, the quantitative gap lemma, profiles of four or more pieces beyond the reduction branch – only matter for pinning the nested threshold at exactly T, not for Conjecture 20 itself).

Open Problem 24. Can a placement rule with access to the full input attain the lex-max set with polynomially many calls to a sibling-packing oracle, or is there a complexity-theoretic obstruction?

Open Problem 25. Characterize the divergence band of Section 9 exactly; determine the approximability of the cardinality objective; decide whether the additive surrogate admits a pseudopolynomial algorithm for integer radii; and settle the complexity of deciding packability of given circles into a *disk* container (NP-hardness is only published for the square).

Acknowledgements

The results of Sections 6–8 were developed with extensive computer assistance (Anthropic's Claude) under a draft–verify workflow: every result was independently rederived from its bare statement by an adversarial verification pass before being accepted, and several statements were corrected or sharpened by that pass; the reports are part of the repository. To be explicit about epistemic status: the mathematical guarantee for every claim is the written proof and, where a proof delegates an identity, an exact symbolic computation; the verification workflow and the numerical sweeps are quality control and evidence, never a substitute for proof.

A Verification map

Every theorem of Sections 6–8 is backed in the accompanying repository by (i) a self-contained draft with complete proofs, (ii) a verification script whose blocks separate exact symbolic identities (`sympy`) from numerical sweeps, all green, and (iii) an *adversarial verification report*: an independent rederivation of each result from its statement alone, followed by an audit and directed attacks. The correspondence, as "verified claims (script blocks)", is: Theorem 17 – `suelo_rigido`, 10/10 claims (7 blocks); slack pans and the universal trio frontier – `universal` (5/5 blocks); the κ identity – `h1`, 6/6 claims (5/5 blocks); the width curve and Theorem 21 – `grosor` 8/8 claims, `esquina` 6/6 claims (5/5 blocks); profile thresholds – `perfil_tres`, `cuatro` (`cuatrok` 5/5 blocks); the corona criterion – `corona` (5/5 blocks); occupant walls, the opposite-disk lemma and the metallic floors – `ocupantes` (5/5), `bloqueadores` (6/6); the double-pocket wall and the golden line – `bolsillo` (6/6); three-piece profiles and the golden deformation – `striple` (5/5); the golden counterexample family, Theorem 19 – `aureo` (5/5); the pan-exchange golden floor – `batalla2` (5/5). Several statements were corrected or sharpened by the adversarial round (a missing hypothesis in the trio frontier, a refuted "leak" family, the $\alpha \geq o_1$ case split); the reports record each incident.

B Complete proof of Theorem 17

Throughout, work in the normalization of Theorem 17: $r_1 = 1$, $t = r_2$, $u = r_3$, $v = r_4$, width $\omega > 0$, pan radius $R = 1 + t$, and recall $b(t) = t(1+t)/(1+t+t^2)$, $T^3 = T^2 + T + 1$, $t^* = 1/T$ (equivalently $t^{*3} + t^{*2} + t^* = 1$), $\varphi = (1+\sqrt{5})/2$. All exact algebraic identities used below are additionally verified in `code/rigido.py` (block V5).

Lemma 26 (Half-angle identity)**.** *Let two circles of radii a, b be internally tangent to the wall of a disk of radius R (centers at distances $R-a$, $R-b$ from the center). If $a + b \leq R$, the minimal angular separation $\theta(a,b)$ guaranteeing disjoint interiors is well defined, lies in $[0,\pi]$, satisfies*

$$\sin^2 \tfrac{\theta(a,b)}{2} = f(a)f(b), \qquad f(x) = \frac{x}{R-x},$$

and is increasing in each argument; two wall-tangent circles at angular separation γ have disjoint interiors iff $\gamma \geq \theta(a,b)$.

Proof. By the law of cosines the center distance $D(\gamma)^2 = (R-a)^2 + (R-b)^2 - 2(R-a)(R-b)\cos\gamma$ is increasing on $[0,\pi]$, and disjointness is $D \geq a+b$. At the threshold, $1 - \cos\theta = \big[(a+b)^2 - (a-b)^2\big]/\big(2(R-a)(R-b)\big) = 2ab/\big((R-a)(R-b)\big)$, i.e. $\sin^2(\theta/2) = f(a)f(b)$; and $f(a)f(b) \leq 1$ is exactly $a + b \leq R$. Monotonicity is that of f. $\square$

Lemma 27 (Construction)**.** *Let $v \leq u \leq t < 1$ and $R = 1+t$. If $\theta(1,u) + \theta(1,v) + \theta(u,v) \leq 2\pi$, then $\{1, u, v\}$ packs in the disk of radius R.*

Proof. All three angles are defined ($1 + u \leq R \iff u \leq t$, and $u + v \leq 2t < R$). Place the three circles wall-tangent with the unit circle in the middle: centers at angles 0, $+\theta(1,u)$, $-\theta(1,v)$. Each circle is contained in the disk (internal wall tangency). The pairs $(1,u)$ and $(1,v)$ are separated by exactly their thresholds. For (u,v), the angular difference is $\Delta = \theta(1,u) + \theta(1,v)$ and the separation is $\gamma = \min(\Delta, 2\pi - \Delta)$. If $\gamma = \Delta$: by monotonicity (Lemma 26, $1 \geq u$), $\theta(1,v) \geq \theta(u,v)$, so $\Delta \geq \theta(u,v)$. If $\gamma = 2\pi - \Delta$: the hypothesis gives $2\pi - \Delta \geq \theta(u,v)$. $\square$

Lemma 28 (Algebraic reduction)**.** *Define $\psi(x) = \sqrt{(t-x)/x}$ (decreasing, $\psi(t) = 0$) and $\tau = t/\sqrt{1+t}$; note the exact identities $\psi(b(t)) = \tau$ and $t/(1+\tau^2) = b(t)$. If $v \leq u \leq t < 1$ and $\psi(u) + \psi(v) \geq \tau$, then $\theta(1,u) + \theta(1,v) + \theta(u,v) \leq 2\pi$, and hence $\{1, u, v\}$ packs in the disk $R = 1 + t$ by Lemma 27.*

Proof. Let $A = \sqrt{f(1)f(u)}$, $B = \sqrt{f(1)f(v)}$, $C = \sqrt{f(u)f(v)}$ (the sines of the half-angles); the goal is $\arcsin A + \arcsin B + \arcsin C \le \pi$. If $\arcsin A + \arcsin B \le \pi/2$ there is nothing to prove, since $\arcsin C \le \pi/2$. Otherwise the goal is equivalent, both sides lying in $[0, \pi/2]$, to $C \le \sin(\arcsin A + \arcsin B) = A\sqrt{1-B^2} + B\sqrt{1-A^2}$. Dividing by $C > 0$ and writing $G(x) = f(1)/f(x) - f(1)^2$,

$$\frac{A\sqrt{1-B^2}}{C} = \sqrt{G(v)}, \qquad \frac{B\sqrt{1-A^2}}{C} = \sqrt{G(u)}, \qquad G(x) = \frac{(1+t)(t-x)}{t^2 x},$$

the last identity by direct computation with $f(1) = 1/t$, $f(x) = x/(1+t-x)$; the radicals are real because $f(1)f(x) \le 1 \iff x \le t$. Thus $\sqrt{G(x)} = \big(\sqrt{1+t}/t\big)\psi(x)$ and, by hypothesis, $\sqrt{G(u)} + \sqrt{G(v)} = \big(\sqrt{1+t}/t\big)(\psi(u) + \psi(v)) \ge \big(\sqrt{1+t}/t\big)\tau = 1$, which is the required inequality. $\square$

The contrapositive used below: hypothesis (F3) implies $\psi(u) + \psi(v) < \tau$. This is the only direction ever used – the *constructive* one – so no exactness of any angular criterion is assumed anywhere in this appendix.

Lemma 29 (Corner minimization)**.** *Let $0 < v \le u \le t < 1$ with $\psi(u) + \psi(v) < \tau$. Then:* (1) *$u, v > b(t)$;* (2) *if $t \ge 1/\varphi$ then $u + v > 1$;* (3) *if $t < 1/\varphi$ then $u + v > t + b(t)$.*

Proof. (1) Each summand is $< \tau$, ψ is decreasing and $\psi(b(t)) = \tau$. (2) $u + v > 2b(t)$ and $2b(t) - 1 = (t^2 + t - 1)/(t^2 + t + 1) \ge 0$ exactly when $t \ge 1/\varphi$. (3) Substitute $\alpha = \psi(u)$, $\beta = \psi(v)$, so $u = U(\alpha)$, $v = U(\beta)$ with $U(z) = t/(1 + z^2)$ strictly decreasing; the region is $\alpha, \beta \ge 0$, $\alpha + \beta < \tau$. Increasing α to $\tau - \beta$ decreases u, so $u + v > U(\tau - \beta) + U(\beta) =: W(\tau - \beta)$. Now $U''(z) = 2t(3z^2 - 1)/(1 + z^2)^3 \le 0$ for $z \le 1/\sqrt{3}$, and $\tau \le 1/\sqrt{3} \iff 3t^2 - t - 1 \le 0 \iff t \le (1 + \sqrt{13})/6 = 0.7676\ldots$, which covers $t < 1/\varphi = 0.618\ldots$ with room to spare. Hence U is concave on $[0, \tau]$, W is concave, and its minimum over $[0, \tau]$ is at the endpoints: $W \ge W(0) = U(0) + U(\tau) = t + t/(1 + \tau^2) = t + b(t)$, by the identity $t/(1 + \tau^2) = b(t)$. $\square$

Proof of the strict floor in Theorem 17. Let $I \in \mathcal{F}$, normalized. By (F3) and the contrapositive of Lemmas 27–28, $\psi(u) + \psi(v) < \tau$. If $t \ge 1/\varphi$, Lemma 29(2) gives $u + v > 1$, contradicting (F2) ($u + v \le 1 - \omega < 1$); so $t < 1/\varphi$ and Lemma 29(3) gives the blocking pressure $u + v > t + b(t)$. Chaining with the witness pressure (F2),

$$t + b(t) \;<\; u + v \;\le\; 1 - \omega \;<\; 1,$$

and $t + b(t) < 1 \iff t(1 + t) < (1 - t)(1 + t + t^2) = 1 - t^3 \iff t^3 + t^2 + t < 1 \iff t < t^*$ (which also proves that $\mathcal{F}$ forces $t < t^*$). Finally

$$\rho \;\ge\; \rho_2 = \frac{u+v}{t} \;>\; \frac{t + b(t)}{t} \;=\; 1 + \frac{1+t}{1+t+t^2} \;=:\; L(t),$$

and L is strictly decreasing ($L' = -(t^2 + 2t)/(1 + t + t^2)^2 < 0$) with $L(t^*) = 1/t^* = T$ (an identity modulo $t^{*3} + t^{*2} + t^* = 1$). Since $t < t^*$: $\rho > L(t) > L(t^*) = T$. $\square$

It remains to prove exactness of the infimum. Two ingredients: the rigid pocket is exact, and infeasibility survives small perturbations.

Proposition 30 (Exact rigid pocket)**.** *For $0 < v \le t < 1$, the triple $\{1, t, v\}$ packs in the disk of radius $R = 1 + t$ if and only if $v \le b(t)$.*

Proof. Rigidity. Centers satisfy $|c_1| \le t$, $|c_t| \le 1$ and $|c_1 - c_t| \ge 1 + t$; the triangle inequality forces equality throughout, so up to rotation $c_1 = (-t, 0)$, $c_t = (1, 0)$. *Sufficiency.* The pocket circle tangent to the wall and to both has radius $b(t)$; any $v \le b(t)$ placed concentrically inside it works. *Necessity.* Let X be the center of v, $d = |X| > 0$, and γ the angle between X and $(1, 0)$. Disjointness reads $d^2 + t^2 + 2dt \cos\gamma \ge (1 + v)^2$ and $d^2 + 1 - 2d \cos\gamma \ge (t + v)^2$, which bound $\cos\gamma$ from below and above; compatibility of the two bounds requires $(1 + t)d^2 \ge (1 + v)^2 + t(t + v)^2 - t(1 + t)$, and with $d \le R - v = 1 + t - v$:

$$(1 + t)(1 + t - v)^2 - (1 + v)^2 - t(t + v)^2 + t(1 + t) \ \ge\ 0.$$

The left-hand side factors *exactly* as $4\,(t^2 + t + 1)\,\big(b(t) - v\big)$ (symbolic identity, V5), whence $v \le b(t)$. □

Lemma 31 (Closure and monotonicity of packability)**.** *Let $D = \{(t, u, v) : 0 < v \le u \le t < 1\}$ and say $p = (t, u, v)$* packs *if there are centers with $|c_1| \le t$, $|c_2| \le 1 + t - u$, $|c_3| \le 1 + t - v$, $|c_1 - c_2| \ge 1 + u$, $|c_1 - c_3| \ge 1 + v$, $|c_2 - c_3| \ge u + v$. Let $E \subseteq D$ be the packable set. Then: (1) E is downward closed in (u, v); (2) E is closed in D; (3) for fixed t, v, the set $\{u \in [v, t] : (t, u, v) \in E\}$ is empty or a closed interval $[v, u_{\max}]$ with the maximum attained; in particular, if $(t, t, v) \notin E$, then $(t, u, v) \notin E$ for all $u \in (u_{\max}, t]$ – the infeasibility of the rigid corner propagates to $\delta := t - u \in [0, \delta_0)$, $\delta_0 = t - u_{\max} > 0$.*

Proof. (1) The same centers witness (all six constraints relax). (2) Witness centers live in a fixed compact ball; take a convergent subsequence of witnesses along $p_k \to p$ and pass to the limit in the six non-strict continuous constraints. (3) By (1) the set is an initial segment, by (2) it is closed. □

Proposition 32 (The infimum is T and is not attained)**.** $\inf\{\rho(I) : I \in \mathcal{F}\} = T$.

Proof. $\ge$ and non-attainment: the strict floor above. $\le$: fix $t < t^*$, set $\varepsilon_t = (1 - t - b(t))/2 > 0$ and $v_t = b(t) + \min\big(\varepsilon_t, (t - b(t))/2\big)$. By Proposition 30, (t, t, v_t) is infeasible; by Lemma 31(3) there is $\delta \in (0, \min(\varepsilon_t, t - v_t))$ with $(t, t - \delta, v_t)$ infeasible. The instance $I_t = (1,\, t,\, t - \delta,\, v_t)$ with $\omega = 1 - (t - \delta + v_t) > 0$ satisfies (F1)–(F3) and the strict orderings, so $I_t \in \mathcal{F}$, and $\rho(I_t) = \rho_2 \le \big(t + b(t) + \varepsilon_t\big)/t = L(t) + \varepsilon_t/t \longrightarrow L(t^*) = T$ as $t \to t^*$ (the other tails are smaller: $\rho_1 = t + u + v < 3t < T$ and $\rho_3 = v/u < 1$). □

Proof of Remark 18. A packing in a disk of radius $R' \le R$ is a packing in the disk of radius R (translate to concentric position; containment is preserved, pairwise constraints are unchanged). Hence non-packability at radius R implies non-packability at radius $r_1 + r_2 \le R$, so an instance with a slack pan satisfying (F2) and (F3) at its own radius satisfies (F2)–(F3) at radius $r_1 + r_2$, and the floor applies verbatim; the approximating family of Proposition 32 has $R = r_1 + r_2$, which is admissible, so the infimum is unchanged. □

C Complete proofs for the width program

This appendix proves every statement of Section 7. Throughout, normalize $r_m = 1$ and $\omega = w/r_m \in (0, 1)$, and write $\beta := 1 - \omega$. The reinsertion resources of the exchange step are the two root nests $A = N(1)$ (the vacated disk D_m) and $B = N(\beta)$ (the hole H_m), plus recursive nesting inside S itself: a ring x fits in the hole of a ring y iff $x \le y - \omega$. For siblings in a disk of capacity c we use: one circle, $s \le c$; two circles, $s + s' \le c$, which is *exact* (sufficiency by tangent placement on a diameter; necessity from $s + s' \le |c_1 - c_2| \le |c_1| + |c_2| \le (c - s) + (c - s')$); and for three or more only *constructive* certificates – the row of Lemma 3 and explicit placements – so that every lower bound below is unconditional. For a profile $S = \{s_1 \ge \cdots \ge s_k\} \subset (0, 1)$,

$$\rho_{\text{needed}}(S) = \max\Big(\sum S,\ \max_j \big(\textstyle\sum_{l>j} s_l\big)/s_j\Big),$$

and $\rho^*_k(\omega) = \inf\{\rho_{\text{needed}}(S) : S \text{ not reinsertable in } A \uplus B\}$. Two elementary facts are used without mention: ρ_{needed} is monotone under adding smaller rings, and if a prefix $\{s_1, \ldots, s_j\}$ is already non-reinsertable then so is S (every placement restricts).

C.1 Profile thresholds

Proposition 33 (Two rings)**.** *$S = \{\sigma_1 \ge \sigma_2\}$ is non-reinsertable iff $\sigma_1 + \sigma_2 > 1$ and $\sigma_2 > 1 - \omega$. Hence $\rho^*_2(\omega) = \max\big(1, 2(1-\omega)\big)$, approached by $\sigma_1 = \sigma_2 \to \max(\frac{1}{2}, 1-\omega)^+$.*

Proof. The possible placements are: both in A (iff $\sigma_1 + \sigma_2 \le 1$, two-circle exact); one in each nest (the one in B measures $\ge \sigma_2$, so it requires $\sigma_2 \le \beta$); σ_2 nested in σ_1 (requires $\sigma_2 \le \sigma_1 - \omega \le \beta$); both in B (stronger than one in each). Both stated conditions kill all four; failure of either condition enables the first or second placement. The infimum: blocking forces $\rho \ge \sigma_1 + \sigma_2 > \max(1, 2\beta)$, and the twin family attains the bound in the limit (its tail ratios tend to 1 and 0). □

Proposition 34 (Three rings: characterization)**.** *Let $s_1 \ge s_2 \ge s_3$ in $(0, 1)$. S is non-reinsertable iff exactly one of the following holds (the cases are disjoint by the position of s_1, s_2 relative to β):*

- *(i) $s_2 > \beta$ and $s_1 + s_2 > 1$ (then s_3 is arbitrary);*
- *(ii) $s_2 > \beta$, $s_1 + s_2 \le 1$, $s_3 > \beta$, and the trio does not pack in the unit disk (possible only for $\omega > \frac{1}{2}$);*
- *(iii) $s_2 \le \beta < s_1$, $s_1 + s_3 > 1$, $s_2 + s_3 > \beta$, and $s_3 > s_1 - \omega$;*
- *(iv) $s_1 \le \beta$, $s_2 + s_3 > 1$, and $s_3 > s_1 - \omega$.*

Proof. The nesting forests on three rings are finite: no edge; one edge (s_3 in s_1, s_3 in s_2, s_2 in s_1); the chain $s_3 \subset s_2 \subset s_1$; the star $\{s_2, s_3\}$ in the hole of s_1. Top-level rings are distributed among A and B under the sibling rules above; S is reinsertable iff some combination is feasible.

(i). Neither s_1 nor s_2 fits in B ($> \beta$) and s_2 cannot nest in s_1 ($s_2 \le s_1 - \omega < \beta < s_2$ is absurd), so both must go to A, and $s_1 + s_2 > 1$ forbids it: blocked for every s_3. Conversely, with $s_2 > \beta$ and $s_1 + s_2 \le 1$: if $s_3 \le \beta$, place $\{s_1, s_2\}$ in a row in A and s_3 in B; if $s_3 > \beta$, all three live in the band $(\beta, 1)$, B is useless, nobody nests ($s_i - \omega < \beta < s_3$), so all three must go to A and reinsertability is exactly packability of the trio in the unit disk – case (ii), whose condition is also sufficient by the same argument. Habitability of (ii): $s_1 + s_2 > 2\beta$ and $s_1 + s_2 \le 1$ force $\omega > \frac{1}{2}$.

(iii). ($\Leftarrow$) $s_1 > \beta$ has no parent and does not fit in B: $s_1 \in A$. No edge survives: $s_3 > s_1 - \omega \ge s_2 - \omega$ kills all three. A companion of s_1 in A needs $s_1 + s_i \ge s_1 + s_3 > 1$: impossible. So s_2, s_3 must both go to B, and $s_2 + s_3 > \beta$ forbids it. ($\Rightarrow$) If $s_1 + s_3 \le 1$: $\{s_1, s_3\}$ in A, s_2 in B. If $s_2 + s_3 \le \beta$: $\{s_2, s_3\}$ in B, s_1 in A. If $s_3 \le s_1 - \omega$: nest s_3 in s_1, $s_1 \to A$, $s_2 \to B$.

(iv). ($\Leftarrow$) $s_2 + s_3 > 1$ makes *every* pair infeasible in any root ($s_i + s_j \ge s_2 + s_3 > 1 > \beta$), so each root hosts at most one top-level ring; two roots cannot host three rings unless someone

nests, and $s_3 > s_1 - \omega$ kills all edges as in (iii). ($\Rightarrow$) If $s_2 + s_3 \le 1$: $\{s_2, s_3\}$ in A, s_1 in B ($s_1 \le \beta$). If $s_3 \le s_1 - \omega$: nest s_3 in s_1, $s_1 \to A$, $s_2 \to B$. $\square$

Theorem 35 (Three rings: closed formula)**.** *For all $\omega \in (0,1)$,*

$$\rho_3^*(\omega) = \max\Big(1,\ \min\big(2(1-\omega),\ \max\big(\varphi,\ \tfrac{2}{1+2\omega}\big)\big)\Big),$$

with extremal families (as $\varepsilon \to 0^+$): $\{\frac{1}{2}+\omega, \frac{1}{2}+\varepsilon, \frac{1}{2}+\varepsilon\}$ on the hyperbola, $\{1/\varphi, \frac{1}{2}+\varepsilon, \frac{1}{2}+\varepsilon\}$ on the golden plateau $\omega \in \big[\frac{\sqrt{5}-2}{2}, 1-\frac{\varphi}{2}\big]$, and $\{\beta+\varepsilon, \beta+\varepsilon, \delta\}$ beyond. The proof is purely additive: neither case (ii) nor any three-circle geometry enters.

Proof. We bound the infimum of ρ_{needed} over each case of Proposition 34 and take the minimum.

Case (i). $\rho \ge s_1 + s_2 > \max(1, 2\beta)$, and the family $s_1 = s_2 = \max(\beta, \frac{1}{2}) + \varepsilon$, $s_3 = \delta \to 0$ is blocked with $\rho_{\text{needed}} \to \max(1, 2\beta) = \rho_2^*$. (This "pair plus dust" argument also shows $\rho_{k+1}^* \le \rho_k^*$ for every k.)

Case (ii). Any blocked profile has $\sum S > 1$ (otherwise the row of Lemma 3 places the trio in A), so its infimum is ≥ 1, the formula's value on $\omega \ge \frac{1}{2}$.

Case (iii). From $s_3 > \max(1-s_1,\, s_1-\omega)$ and $s_2 \ge s_3$: $\sum S \ge g(s_1) := s_1 + 2\max(1-s_1, s_1-\omega)$, which decreases until $s_1 = (1+\omega)/2$ and increases after, with minimum $(3-\omega)/2$; and $s_1 > \beta$. If $\omega \le \frac{1}{3}$ then $\beta \ge (1+\omega)/2$ and $\sum S > g(\beta) = 3 - 5\omega \ge 2 - 2\omega$; if $\omega \in (\frac{1}{3}, \frac{1}{2})$, $\sum S > (3-\omega)/2 \ge 2 - 2\omega$; if $\omega \ge \frac{1}{2}$, $\sum S \ge s_1 + s_3 > 1$. In all ranges case (iii) is dominated by the formula's value: it never furnishes the minimum.

Case (iv). Set $q = s_2 + s_3$. Blocking gives $q > 1$ and $q \ge 2s_3 > 2(s_1 - \omega)$, with $s_1 \le \beta$, while $\rho \ge \max(s_1 + q,\ q/s_1)$. For fixed s_1 both bounds grow with q, so $q \downarrow Q(s_1) := \max(1, 2(s_1-\omega))$. On the branch $s_1 \ge \frac{1}{2} + \omega$ ($Q = 2(s_1 - \omega)$) both $3s_1 - 2\omega$ and $2 - 2\omega/s_1$ increase in s_1: the branch minimum is $\max\big(\frac{3}{2}+\omega, \frac{2}{1+2\omega}\big)$ at the left endpoint. On the branch $s_1 \le \frac{1}{2} + \omega$ ($Q = 1$), $h(s_1) = \max(s_1 + 1, 1/s_1)$ decreases until the crossing $s_1^2 + s_1 = 1$, i.e. $s_1 = 1/\varphi$ where $h = \varphi$, and increases after; the branch minimum sits at $s_1^* = \min(\beta, \frac{1}{2}+\omega, 1/\varphi)$. Comparing (exact crossings): for $\omega \le \omega_1 := \frac{\sqrt{5}-2}{2}$ the infimum is $\frac{2}{1+2\omega}$ (the crossing $\frac{2}{1+2\omega} = \frac{3}{2} + \omega$ is $2\omega^2 + 4\omega - \frac{1}{2} = 0$, positive root ω_1); for $\omega_1 \le \omega \le 1 - 1/\varphi$ it is φ; for $1 - 1/\varphi \le \omega < \frac{1}{2}$ it is $\max(2-\omega, 1/\beta) = 1/\beta \ge 2\beta$: dominated by case (i). For $\omega \ge \frac{1}{2}$ case (iv) is empty ($s_2 + s_3 \le 2\beta < 1$). The lower bounds are approached by the stated families ($s_2 = s_3 = q/2$ with s_1 at each branch minimum satisfies all blocking conditions with margin ε).

Global minimum. $\frac{2}{1+2\omega} < 2(1-\omega)$ on $(0, \frac{1}{2})$ (equivalent to $\omega(1-2\omega) > 0$) and $\varphi \le 2(1-\omega) \iff \omega \le 1 - \varphi/2$: the case-wise minimum is the stated formula. $\square$

Proposition 36 (All profile sizes)**.** *$\rho_k^*(\omega) = \rho_3^*(\omega)$ for every $k \ge 3$ and every $\omega \in (0,1)$.*

Proof. ($\le$) Dust: append $k-3$ rings $\delta \to 0$ to any blocked three-ring witness. ($\ge$) Let S be non-reinsertable with k rings and $\rho := \rho_{\text{needed}}(S)$.

Step 0. The row of all of S in A fails: $\sum S > 1$, so $\rho > 1$; this closes $\omega \ge \frac{1}{2}$. Assume $\omega < \frac{1}{2}$ and $\rho < 2\beta$ (otherwise done).

Step 1. Two rings $> \beta$ would give $\sum S > 2\beta \ge \rho \ge \sum S$. If only $s_1 > \beta$: the placement "$s_1 \to A$ alone, the rest in a row in B" fails, and since each $s_i \le \beta$ fits alone the failure is the sum: $\sum S - s_1 > \beta$, whence $\sum S > 2\beta$: impossible. So $s_1 \le \beta$.

Step 2. "$s_1 \to B$, the rest in a row in A" fails: $Q := \sum S - s_1 > 1$. From $\rho \ge s_1 + Q$ and $\rho \ge Q/s_1$: $\rho \ge \min_{s \le \beta} \max(1+s, 1/s)$, which equals φ if $\beta \ge 1/\varphi$ and $1/\beta$ otherwise; in either case $\rho \ge \rho_3^*$ for all $\omega \ge \frac{\sqrt{5}-2}{2}$ (on the plateau $\rho_3^* \le \varphi$; beyond it $\rho_3^* = 2\beta \le \varphi$ resp. $2\beta \le 1/\beta$ for $\beta \le 1/\sqrt{2}$). Assume then $\omega < \frac{\sqrt{5}-2}{2}$, where $\rho_3^* = \frac{2}{1+2\omega}$, and towards a contradiction $\rho < \frac{2}{1+2\omega}$. From $\rho \ge Q/s_1 > 1/s_1$ we get $s_1 > \frac{1}{2} + \omega$, in particular $s_1 - \omega > \frac{1}{2}$.

Step 3. Let $p := \#\{i : s_i > s_1 - \omega\}$; all such rings exceed $\frac{1}{2}$ and $\sum S \le \rho < 2$ allows at most three, so $p \in \{1, 2, 3\}$.

$p \ge 3$: $s_1, s_2, s_3 > s_1 - \omega > \frac{1}{2}$. No pair fits together in a root (sums $> 1 > \beta$, two-circle exact) and nobody nests ($s_j \le s_i - \omega \le s_1 - \omega < s_j$ is absurd): the prefix $\{s_1, s_2, s_3\}$ is already non-reinsertable, so $\rho \ge \rho_{\text{needed}}(\{s_1, s_2, s_3\}) \ge \rho_3^*$.

$p = 2$: $s_3, \dots, s_k \le s_1 - \omega$. The placement "$\{s_3, \dots, s_k\}$ in a row inside the hole of s_1 (capacity $s_1 - \omega$), $s_1 \to A$, $s_2 \to B$" fails, and the only piece that can fail is the row: $R := \sum_{i\ge3} s_i > s_1 - \omega$. With $s_2 > s_1 - \omega$, the tail $Q = s_2 + R > 2(s_1 - \omega)$ and $Q > 1$ (Step 2): exactly the case-(iv) program of Theorem 35, whose value is $\ge \rho_3^*$. (This is the only biting branch: the dust of $(\le)$ lives here.)

$p = 1$: $s_2 \le s_1 - \omega$. The placement "s_2 in the hole of s_1, $s_1 \to B$, $\{s_3, \dots, s_k\} \to A$ in a row" fails: $R' := \sum_{i\ge3} s_i > 1$. Then $\sum S = s_1 + s_2 + R' > (s_2 + \omega) + s_2 + 1 = 1 + 2s_2 + \omega$ and the tail of s_2 gives $\rho \ge R'/s_2 > 1/s_2$, so

$$\rho \;\ge\; \min_s \max\bigl(1 + 2s + \omega,\ 1/s\bigr) = \tfrac{(1+\omega)+\sqrt{(1+\omega)^2+8}}{2} \;\ge\; 2,$$

contradicting $\rho < 2\beta < 2$. □

Corollary 37 (Exact combinatorial threshold)**.** *For $\omega < \omega_T = \frac{1}{T} - \frac{1}{2}$ no profile of any size blocks the exchange step with $\rho < T$; for $\omega > \omega_T$ the witness $\{\frac{1}{2} + \omega, \frac{1}{2} + \varepsilon, \frac{1}{2} + \varepsilon\}$ blocks with $\rho = \frac{2}{1+2\omega} < T$.* □

C.2 The blocking frontier of the canonical trio

Work in the canonical template: the trio $\{\alpha, \sigma_1, \sigma_2\}$ with $0 < \sigma_2 \le \sigma_1 \le 1 < \alpha$ must pack in the pan of radius $R = \alpha + 1$. With the half-angle machinery of Appendix B (Lemma 26), set $f(x) = x/(R - x)$, $\theta(a, b) = 2 \arcsin \sqrt{f(a)f(b)}$ and $F(\sigma_1, \sigma_2) = \theta(\alpha, \sigma_1) + \theta(\alpha, \sigma_2) + \theta(\sigma_1, \sigma_2)$; by the constructive direction (Lemma 27), $F \le 2\pi$ implies the trio packs, so non-packability implies $F \ge 2\pi$ on the closure. Each θ is strictly increasing in both arguments, so $F = 2\pi$ defines the *blocking frontier* $\sigma_2 = h(\alpha, \sigma_1)$ where it meets the band. Two evaluations used repeatedly: $f(\alpha) = \alpha$ (since $R - \alpha = 1$) and $1 - \alpha f(\sigma) = R(1 - \sigma)/(R - \sigma)$.

Lemma 38 (Closed frontier)**.** *Let $\alpha > 1$ and $\sigma := 1/\sqrt{\alpha(\alpha + 1)}$. In the band $0 < \sigma_2 \le \sigma_1 \le 1$,*

$$F(\sigma_1, \sigma_2) = 2\pi \iff t(\sigma_1) + t(\sigma_2) = \sigma, \qquad t(s) = \sqrt{\tfrac{1-s}{s}},$$

and $\sigma = t(b(\alpha))$. Consequently $h(\alpha, \sigma_1) = t^{-1}(\sigma - t(\sigma_1))$ with $t^{-1}(u) = 1/(1 + u^2)$, defined and strictly decreasing exactly for $\sigma_1 \in [s^*, 1]$ where $s^* = 4\alpha(\alpha + 1)/(2\alpha + 1)^2$ is the diagonal point ($t(s^*) = \sigma/2$); the range of h is $[b(\alpha), s^*]$, with $h(\alpha, 1) = b(\alpha)$.

Proof. On a frontier point all three angles are interior: for the pairs (α, σ_i), $P = \alpha f(\sigma_i) < 1$ since $\sigma_i < 1$ (and $\alpha f(1) = 1$); for (σ_1, σ_2), $\sigma_1 + \sigma_2 \le 2 < R$. Write $A = \theta(\alpha, \sigma_1)/2$, $B = \theta(\alpha, \sigma_2)/2 \in (0, \pi/2)$, so that $\sin^2 A = \alpha f(\sigma_1)$, $\cos^2 A = R(1 - \sigma_1)/(R - \sigma_1)$ (same for B with σ_2), and on the frontier $\theta(\sigma_1, \sigma_2)/2 = \pi - A - B \in (0, \pi)$.

($\Rightarrow$) Taking sines, $\sin\bigl(\theta(\sigma_1,\sigma_2)/2\bigr) = \sin(A+B) = \sin A\cos B + \cos A \sin B$. The cross terms factor:

$$(\sin A\cos B)^2 = \alpha f(\sigma_1)\cdot\frac{R(1-\sigma_2)}{R-\sigma_2} = \alpha R\, t(\sigma_2)^2\, f(\sigma_1)f(\sigma_2),$$

because $(1-\sigma_2)/(R-\sigma_2) = t(\sigma_2)^2 f(\sigma_2)$; symmetrically with $t(\sigma_1)$. Since $\sin(\theta(\sigma_1,\sigma_2)/2) = \sqrt{f(\sigma_1)f(\sigma_2)} \neq 0$, dividing gives $1 = \sqrt{\alpha R}\,\bigl(t(\sigma_1)+t(\sigma_2)\bigr)$, the stated equation with $\sigma = 1/\sqrt{\alpha R}$. The identity $(1-b)/b = 1/(\alpha(\alpha+1))$ for $b = b(\alpha)$ gives $t(b(\alpha)) = \sigma$.

($\Leftarrow$) For fixed σ_1, both $F(\sigma_1,\cdot)$ (increasing) and $t(\sigma_1)+t(\cdot)$ (decreasing) are strictly monotone in σ_2; by ($\Rightarrow$) their level sets coincide where the frontier meets the band, and by monotonicity each has a unique solution. Existence: on the diagonal, $F(s,s)$ increases from 0^+ to $F(1,1) > 2\pi$, so there is a unique s° with $F = 2\pi$, and ($\Rightarrow$) forces $t(s^\circ) = \sigma/2$, i.e. $s^\circ = s^*$. For $\sigma_1 \in [s^*, 1]$, $F(\sigma_1, 0^+) \le \pi < 2\pi \le F(\sigma_1,\sigma_1)$: the frontier crosses, and $t(\sigma_1) \le \sigma/2$ keeps $h(\sigma_1) \le \sigma_1$. For $\sigma_1 < s^*$ it would need $\sigma_2 > \sigma_1$. Endpoints: $h(s^*) = s^*$ and $h(1) = t^{-1}(\sigma) = b(\alpha)$. $\square$

Theorem 39 (The κ identity)**.** *On the frontier, for every $\alpha > 1$ and $\sigma_1 < 1$,*

$$\kappa := -\frac{\partial h}{\partial \sigma_1} = \sqrt{\frac{g(\sigma_2)}{g(\sigma_1)}}, \qquad g(s) = s^3(1-s), \qquad \textit{and} \qquad \kappa \ge 1,$$

with equality iff $\sigma_1 = \sigma_2 = s^*$.

Proof. The identity: $t'(s) = -1/(2\sqrt{g(s)})$, and differentiating $t(\sigma_1) + t(h(\sigma_1)) = \sigma$ gives $\kappa = t'(\sigma_1)/t'(\sigma_2) = \sqrt{g(\sigma_2)/g(\sigma_1)}$.

The inequality, in the t-coordinate: since t reverses order, $\sigma_2 \le \sigma_1 \iff t_2 \ge t_1$ on the segment $t_1 + t_2 = \sigma$, and the claim is $G(t_2) \ge G(t_1)$ for

$$G(t) := g\bigl(1/(1+t^2)\bigr) = \frac{t^2}{(1+t^2)^4}, \qquad G'(t) = \frac{2t(1-3t^2)}{(1+t^2)^5}:$$

G increases on $[0, 1/\sqrt{3}]$ and decreases after. Note $\alpha > 1 \iff \alpha(\alpha+1) > 2 \iff \sigma < 1/\sqrt{2}$. If $\sigma \le 1/\sqrt{3}$, both $t_i \le \sigma$ lie in the increasing range and $t_2 \ge t_1$ gives the claim, with equality only at $t_1 = t_2$. If $\sigma > 1/\sqrt{3}$ and $t_2 > 1/\sqrt{3}$ (otherwise the previous case applies), then $t_1 = \sigma - t_2 < 1/\sqrt{2} - 1/\sqrt{3}$, so $G(t_1) \le G(1/\sqrt{2} - 1/\sqrt{3}) = 0.01574\ldots$, while $t_2 \in (1/\sqrt{3}, \sigma)$ lies in the decreasing range, so $G(t_2) > G(\sigma) > G(1/\sqrt{2}) = \frac{8}{81}$: here $\kappa^2 > \frac{8/81}{0.0158} > 6$, far from tight. $\square$

Corollary 40 (Minimum of $\sigma_1+\sigma_2$; Tribonacci closure)**.** *For every $\alpha > 1$, the minimum of $\sigma_1 + \sigma_2$ over the non-packable region $\{F \ge 2\pi,\ 0 < \sigma_2 \le \sigma_1 \le 1\}$ is $1 + b(\alpha)$, attained at $(\sigma_1,\sigma_2) = (1, b(\alpha))$. Consequently a blocking satisfying the witness wall $\sigma_1 + \sigma_2 \le \alpha - \omega$ forces $1 + b(\alpha) \le \alpha - \omega$, i.e. $\alpha \ge T_{1+\omega}$ (the root of $\alpha^3 = (1+\omega)(\alpha^2+\alpha+1)$); in particular no blocking exists for $\alpha \le T$, at any width.*

Proof. The region is non-empty ($F(1,1) > 2\pi$) and a minimum cannot have $F > 2\pi$ (decrease σ_2: the sum drops, F stays $\ge 2\pi$ by continuity until the frontier is crossed, and $F(\sigma_1, 0^+) < 2\pi$ guarantees it is crossed). On the frontier, $d(\sigma_1 + h)/d\sigma_1 = 1 - \kappa \le 0$ (Theorem 39): the minimum sits at $\sigma_1 = 1$ and equals $1 + b(\alpha)$. The sign of $1 + b(\alpha) - (\alpha - \omega)$ is governed by the polynomial identity $1 + b(\alpha) - \alpha = -(\alpha^3 - \alpha^2 - \alpha - 1)/(\alpha^2+\alpha+1)$ and its deformation $b(\alpha) - (\alpha - 1 - \omega) = -\bigl[\alpha^3 - (1+\omega)(\alpha^2+\alpha+1)\bigr]/(\alpha^2+\alpha+1)$, whose numerator has the single positive root $T_{1+\omega}$ (Descartes' rule); $T_1 = T$. $\square$

C.3 Positive width: the exact curve and the $13/7$ corner

The full blocking program of the canonical template adds the remaining resources to the trio wall. With capacities $c_2 := 1-\omega$ (the hole H_m) and $c_4 := \alpha-\omega-1$ (σ_2 back in u next to m, two-circle exact), a realizable blocking satisfies (closures for infima)

$$\text{(B1) trio non-packable,}\quad \text{(B2) } \sigma_2 \ge c_2,\quad \text{(B4) } \sigma_2 \ge c_4,\quad \text{(W) } \sigma_1+\sigma_2 \le \alpha-\omega,$$

plus the band $\sigma_2 \le \sigma_1 \le 1$ and $\alpha-\omega \ge 1$, and $T_{\mathrm{can}}(\omega)$ is the infimum of $\rho = \max\big(\sigma_1+\sigma_2, (1+\sigma_1+\sigma_2)/\alpha\big)$ over blockings. (At $\omega = 0$ the program is empty – (B2) would force $\sigma_2 \ge 1$ – which is why the limit value T uses only (B1)+(W).)

Proposition 41 (Witness branch)**.** *The infimum of ρ over (B1)+(W) alone is $\Phi(\omega) = T_{1+\omega}-\omega$, attained in the limit $\alpha = T_{1+\omega}$, $\sigma_1 \to 1$, $\sigma_2 \to b(\alpha) = \alpha-1-\omega$. Moreover Φ is strictly increasing and concave, with*

$$\Phi'(\omega) = \frac{2\alpha+1}{\alpha^2(\alpha^2+2\alpha+3)}\Big|_{\alpha=T_{1+\omega}} > 0, \qquad \Phi'(0) = \frac{2T+1}{7T^2+4T+3} = 0.1374\ldots,$$

and on $[0,\frac{1}{7}]$ the chord–tangent bounds $T+(13-7T)\,\omega \le \Phi(\omega) \le T+\Phi'(0)\,\omega$ hold, the chord endpoint being exact: $T_{8/7} = 2$ and $\Phi(\frac{1}{7}) = \frac{13}{7}$.

Proof. By Corollary 40, blocking at a given α forces $\rho \ge \sigma_1+\sigma_2 \ge 1+b(\alpha)$ and (W) forces $\alpha \ge T_{1+\omega}$; b is strictly increasing ($b' = (2\alpha+1)/(\alpha^2+\alpha+1)^2$), so the infimum over admissible α is $1+b(T_{1+\omega}) = T_{1+\omega}-\omega$ (the cubic identity with equality), approached by $\sigma_1 = 1-\varepsilon$, $\sigma_2 = b(\alpha)+\varepsilon'$. The derivative: implicit differentiation of $\alpha^3 = (1+\omega)(\alpha^2+\alpha+1)$ and substitution of $1+\omega = \alpha^3/(\alpha^2+\alpha+1)$ reduce the numerator of $d\alpha/d\omega - 1$ to $2\alpha+1$ (symbolic identity). Concavity: $\frac{d}{d\alpha}\log\Phi' = \frac{2}{2\alpha+1} - \frac{2}{\alpha} - \frac{2\alpha+2}{\alpha^2+2\alpha+3} < 0$ and α increases with ω. $T_{8/7} = 2$ because $2^3 = 8 = \frac{8}{7}\cdot 7$. □

Adding (B2) gives the second branch: every full blocking has $\rho \ge \sigma_1+\sigma_2 \ge 2(1-\omega)$. The maximum of the two branches is minimized at their crossing $T_{1+\omega} = 2-\omega$, i.e. at the root $\omega_\times = 0.0754315\ldots$ of $2\omega^3-10\omega^2+14\omega-1$, giving the uniform floor $T_{\mathrm{can}}(\omega) \ge 2(\alpha_\times-1) = T+0.00985\ldots$ for $\omega \in (0, 0.30]$ – already enough for "width only raises the floor". The exact curve is finer. For fixed α, pass to the coordinate $t_i = t(\sigma_i)$, where (B1) becomes *linear* (Lemma 38): $t_1+t_2 \le \Theta(\alpha) := t(b(\alpha))$, and the walls read $t_2 \le \tau := t(c)$, $c := \max(c_2, c_4)$. If $\alpha > 2+\omega$ there is no blocking at all ((B4) and the band give $\sigma_1+\sigma_2 \ge 2c_4 > \alpha-\omega$, violating (W)); so $\alpha-\omega \in [1,2]$ effectively.

Lemma 42 (Conditional minimum)**.** *Let $\alpha \in (1, 2+\omega)$ and $c = \max(1-\omega, \alpha-\omega-1) < 1$. The infimum of $S = \sigma_1+\sigma_2$ over non-packable trios with $\sigma_2 \ge c$ and $\sigma_2 \le \sigma_1 \le 1$ is*

(a) $1+b(\alpha)$ if $c \le b(\alpha)$;   (b) $c+\sigma\big(\Theta(\alpha)-t(c)\big)$ if $b(\alpha) < c \le s^*(\alpha)$;   (c) $2c$ if $c > s^*(\alpha)$,

where $\sigma(u) = 1/(1+u^2)$. Moreover each case is non-decreasing in α on its domain.

Proof. $\sigma(\cdot)$ is decreasing, so S drops when either t_i rises. In the truncated box $\{t_1 \le t_2 \le \tau,\ t_1+t_2 \le \Theta\}$: if $2\tau \le \Theta$ (case (c)) the box corner (τ,τ) is admissible and optimal, $S = 2\sigma(\tau) = 2c$. Otherwise the minimum lives on the segment $t_1+t_2 = \Theta$, where

$$\frac{dS}{dt_2} = -\sigma'(\Theta-t_2)+\sigma'(t_2) = 2\big[\sqrt{G(t_1)}-\sqrt{G(t_2)}\big] \le 0$$

by the G-lemma inside Theorem 39 ($\sigma'(x) = -2x/(1+x^2)^2$, $(\sigma'/2)^2 = G$): push t_2 to its cap. If $\tau \geq \Theta$ (case (a)): $t_2 = \Theta$, $t_1 = 0$, i.e. $(\sigma_1, \sigma_2) = (1, b(\alpha))$. If $\tau < \Theta$ (case (b)): $t_2 = \tau$, $t_1 = \Theta - \tau$. Monotonicity in α: (a) is $1 + b(\alpha)$ with b increasing; (b) with $c = c_2$ fixed has $dS/d\alpha = \sigma'(\Theta - \tau)\,\Theta'(\alpha) \geq 0$ ($\sigma' < 0$, $\Theta' < 0$); (b)/(c) with the moving cap $c = c_4$ ($\alpha \geq 2$, $\tau_4 = t(c_4)$) has $dS/d\alpha = \tau_4'\,(-2)\big[\sqrt{G(\tau_4)} - \sqrt{G(t_1)}\big] + \sigma'(t_1)\,\Theta' \geq 0$, since $\tau_4' < 0$, $G(\tau_4) \geq G(t_1)$ ($\tau_4 \geq t_1$ on the segment, same G-lemma) and the last two factors are both negative. $\square$

Proposition 43 (The exact curve)**.** *For $\omega \in (0, 0.30]$, the proved lower bound for T_{can} is*

$$T_{\mathrm{can}}(\omega) \;\geq\; \begin{cases} 2(1-\omega) & \omega \in (0, \omega_1], \\ \alpha_m(\omega) - \omega & \omega \in [\omega_1, \frac{1}{7}], \\ \Phi(\omega) & \omega \in [\frac{1}{7}, 0.30], \end{cases}$$

where $\omega_1 = 0.0413570\ldots$ is the root in $(\frac{1}{25}, \frac{1}{14})$ of $4\omega^3 - 20\omega^2 + 25\omega - 1$ and $\alpha_m(\omega) \in (1, 2]$ is the unique root of $t(\alpha - 1) + t(1 - \omega) = \Theta(\alpha)$, an algebraic function of degree 6 (its polynomial $P(\alpha, \omega)$, of bidegree $(6, 2)$, is irreducible over $\mathbb{Q}[\alpha, \omega]$). Equality holds on the witness branch and at the corner; on the other two branches the matching upper bound inherits the angular-criterion caveat.

Proof. Fix ω and minimize the value of Lemma 42 over α subject to (W).

H_m branch ($\omega \leq \omega_1$). Case (c) gives $S = 2(1-\omega)$, constant in α, admissible iff (W): $\alpha \geq 2 - \omega$, and iff $c_2 > s^*(\alpha)$, i.e. $\alpha \leq \bar{\alpha}(\omega)$. The window is non-empty iff $s^*(2-\omega) \leq 1 - \omega$, and $(1-\omega) - s^*(2-\omega)$ has numerator $-(4\omega^3 - 20\omega^2 + 25\omega - 1)$; the cubic is strictly increasing on $(0, \frac{5}{6})$ (derivative $(5-2\omega)(5-6\omega)$), so the window exists exactly for $\omega \leq \omega_1$. The other cases do not undercut $2(1-\omega)$ (monotonicity in Lemma 42 plus continuity across case junctions).

Mixed branch ($\omega_1 < \omega \leq \frac{1}{7}$). For $\alpha \leq 2$, case (c) is empty (it would need $1 - \omega \geq s^*(\alpha) \geq s^*(2-\omega)$, against the cubic) and case (a) needs $b(\alpha) \geq 1 - \omega$, impossible ($b(2) = \frac{6}{7} \leq 1 - \omega$). Case (b) with $c = c_2$ has value $S^b(\alpha) = 1 - \omega + \sigma(\Theta - t(1-\omega))$, increasing in α, and (W) is equivalent to $\Psi(\alpha) := \Theta(\alpha) - t(\alpha-1) \geq t(1-\omega)$ with Ψ strictly increasing on $(1, 2]$, $\Psi(1^+) = -\infty$ and $\Psi(2) = t(\frac{6}{7}) \geq t(1-\omega) \iff \omega \leq \frac{1}{7}$: the admissible α form $[\alpha_m, 2]$ and the minimum is $S^b(\alpha_m) = \alpha_m - \omega$ ((W) with equality). For $\alpha \in (2, 2+\omega)$: if $b(\alpha) \leq 1 - \omega$, relax (B4) to (B2) and use monotonicity, $S \geq S^b(\alpha) > S^b(\alpha_m)$; if $b(\alpha) > 1 - \omega$, $S \geq 1 + b(\alpha) > 2 - \omega > \alpha_m - \omega$. Squaring the defining equation twice yields the polynomial P (necessary condition; the right branch is pinned by $\alpha_m \in (1, 2]$), with the junction identities $P(2-\omega, \omega) = (\omega-1)^3(4\omega^3 - 20\omega^2 + 25\omega - 1)$ (so $\alpha_m(\omega_1) = 2 - \omega_1$) and $\alpha_m(\frac{1}{7}) = 2$.

Witness branch ($\omega \geq \frac{1}{7}$). Now $b(T_{1+\omega}) = T_{1+\omega} - 1 - \omega \geq 1 - \omega \iff T_{1+\omega} \geq 2 \iff \omega \geq \frac{1}{7}$: the optimum of case (a) satisfies (B2), (W) forces $\alpha \geq T_{1+\omega}$ (Corollary 40) and the minimum is $\Phi(\omega)$; larger α and the cases (b)/(c_4) are increasing and start at $\Phi(\omega)$; smaller α admit no blocking (for $\alpha \leq 2$, $\Psi(\alpha) \leq \Psi(2) = t(\frac{6}{7}) \leq t(1-\omega)$ with equality only at the corner; case (c) would need $\omega \leq \omega_1$). $\square$

Proof of Theorem 21. Lower bound. Take any blocking at width ω. If $\alpha \geq 2$: by Corollary 40 alone (no (B2), (B4), (W) needed), $\rho \geq 1 + b(\alpha) \geq 1 + b(2) = \frac{13}{7}$. If $\alpha < 2$, use Proposition 43: for $\omega \geq \frac{1}{7}$ there is no blocking with $\alpha < T_{1+\omega}$ and $T_{1+\omega} \geq 2 > \alpha$: empty. For $\omega \leq \omega_1$: $\rho \geq 2(1-\omega) \geq 2(1-\omega_1) > \frac{13}{7}$ (since $\omega_1 < \frac{1}{14}$). For $\omega_1 < \omega < \frac{1}{7}$: $\rho \geq V(\omega) := \alpha_m(\omega) - \omega$, and $V > \frac{13}{7}$ there: if $V(\omega') = \frac{13}{7}$ then $(\alpha, \omega) = (\frac{13}{7} + \omega', \omega')$ satisfies $P = 0$, but

$$P\big(\tfrac{13}{7} + \omega,\ \omega\big) = \frac{4\,(7\omega - 1)\,Q_5(\omega)}{7^6},$$

$$Q_5 = 33614\,\omega^5 + 235298\,\omega^4 + 620830\,\omega^3 + 766066\,\omega^2 + 397831\,\omega - 289,$$

and Q_5 has no root in $[\frac{1}{25}, \frac{1}{7}]$ (exact root isolation; its only positive root is ≈ 0.0007), so the only root of the product there is $\omega = \frac{1}{7}$; since $V(\omega_1) - \frac{13}{7} = \frac{1}{7} - 2\omega_1 > 0$ and V is continuous, $V > \frac{13}{7}$ on all of $[\omega_1, \frac{1}{7})$. Uniqueness of the infimum configuration: equality in the branch $\alpha \geq 2$ forces $\alpha = 2$, $\sigma_1 = 1$, $\sigma_2 = \frac{6}{7}$; then (B2) gives $\omega \geq \frac{1}{7}$ and (W) gives $\omega \leq \frac{1}{7}$.

The approximating family is genuine. At exactly $\alpha = 2$, $\omega = \frac{1}{7}$ there is no family: $\kappa > 1$ gives $\sigma_1 + h(2, \sigma_1) > \frac{13}{7} = \alpha - \omega$ for $\sigma_1 < 1$, violating (W); the corner is approached by *opening* (B4). Fix $\omega = \frac{1}{7}$ and let $\delta \downarrow 0$:

$$\alpha = 2 + \delta, \quad \varepsilon = \delta^2/4, \quad \sigma_1 = 1 - \varepsilon, \quad \sigma_2 = (\alpha - \omega - 1) + \varepsilon/2.$$

All walls hold ((B4) with slack $\varepsilon/2$; (B2) since $\alpha - \omega - 1 \geq 1 - \omega$ for $\alpha \geq 2$; (W) since $\sigma_1 + \sigma_2 = \alpha - \omega - \varepsilon/2$). Genuineness: since $\alpha > T_{8/7} = 2$, Corollary 40 gives $\sigma_2 > b(\alpha)$ strictly, so the trio $\{\alpha, 1, \sigma_2\}$ is non-packable by the *rigidity* of Proposition 30 (the disk $R = \alpha + 1$ is diametrically full; rescale by $t = 1/\alpha$) – not merely by the angular criterion – and the infeasibility propagates from $\sigma_1 = 1$ to $\sigma_1 = 1 - \varepsilon$ by Lemma 31(3) for all small ε. Then $\rho = \alpha - \omega - \varepsilon/2 \to \frac{13}{7}$. □

Remark 44 (Fine structure)**.** The curve is not monotone on $(0, \frac{1}{7}]$: differentiating the defining equation of α_m at the junction and using the exact coincidence of the leading terms ($(\alpha-1)^3(2-\alpha) = (1-\omega_1)^3\omega_1$ at $\alpha = 2 - \omega_1$) gives $V'(\omega_1^+) = c_0/(r' - c_0) > 0$ (numerically $+0.0721$): ω_1 is a *local minimum* of T_{can}, followed by a local maximum at $\omega_{\text{peak}} = 0.0444700$ (the unique root in $(\omega_1, \frac{1}{7})$ – by Sturm – of an explicit degree-8 factor R_8 of the resultant of P and $P_\alpha + P_\omega$), a bump of height $1.1 \cdot 10^{-4}$, before descending to the corner. T_{can} has two local minima, ω_1 and $\frac{1}{7}$, and the global one is the corner.

D Complete proofs for generic containers

This appendix proves every statement of Section 8. Normalize $r_m = 1$, $\omega = w/r_m$. The template: v is the pan, of radius R, containing $\{\alpha\} \cup O \cup \{m\}$ with $O = \{o_1 \geq \cdots \geq o_j\}$, $j \geq 0$, $o_i \geq m = 1$ (rings larger than m agree in F and P); u is the hole of α, capacity $\alpha - \omega \geq 1$; the witness placed $S = \{\sigma_1 \geq \sigma_2\}$ in u, $\sigma_2 \leq \sigma_1 \leq 1$. Holes may be occupied arbitrarily: contents of a hole at any depth are parametrized by $X_y := \sum \text{children}(y)$. The exchange builds a placement P' that preserves the container assignments of all rings $\geq m$; *positions* inside each container are existential (a placement is a nesting forest whose sibling groups pack). A *node* is any ring of radius ≥ 1 that is an occupant of v other than α *and* m, or is nested (at any depth) inside one; nodes and their descendants live outside m. Unless stated otherwise, blocked-exchange walls are closures of strict inequalities, as in Appendix C.

D.1 The universal frontier and the corona criterion

Lemma 45 (Universal frontier)**.** *Let $\{A, x, y\}$ be wall-tangent in a disk of arbitrary radius R, in the interior domain $A + x, A + y, x + y < R$. Set $c := R - A$, $T_c(x) := \sqrt{(c - x)/x}$ and $\tau_R := c/\sqrt{AR}$. If $A \geq \min(x, y)$, then*

$$F = \theta(A, x) + \theta(A, y) + \theta(x, y) \;\geq\; 2\pi \iff T_c(x) + T_c(y) \;\leq\; \tau_R.$$

The direction ($\Rightarrow$) is unconditional; the converse fails without the hypothesis ($R = 1$, $A = 0.01$, $x = y = 0.45$ is a counterexample).

Proof. The cross term factors for arbitrary R: with $\sin^2(\theta_{Az}/2) = f(A)f(z)$, $f(z) = z/(R-z)$, one has the identity

$$f(A)f(x)\big(1 - f(A)f(y)\big) = \frac{AR}{(R-A)^2} \cdot \frac{c-y}{y} \cdot f(x)f(y),$$

so $\sin(\theta_{Ax}/2)\cos(\theta_{Ay}/2) = \frac{\sqrt{AR}}{c}\, T_c(y)\sqrt{f(x)f(y)}$ and, adding the symmetric term, $\sin s = \frac{\sqrt{AR}}{c}\big(T_c(x)+T_c(y)\big)\sin w$ with $s = (\theta_{Ax}+\theta_{Ay})/2$, $w = \theta_{xy}/2$. The linear form is thus $\sin s \le \sin w$, while $F \ge 2\pi$ is $s+w \ge \pi$. Since $w \le \pi/2$: $s+w \ge \pi \Rightarrow \sin s \le \sin w$ always; the converse needs $s > w$, which the hypothesis provides: if $A \ge y$, monotonicity gives $\theta(A,x) \ge \theta(y,x) = 2w$, so $s > w$. ◻

Uniform companions (same computations as in Appendix C, now for general R): the general pocket $b_R(A) = ARc/(AR + c^2)$, equal to Descartes' b_2 when $R = A + B$, increasing in R and always $\le A$; and the slope identity $\kappa = \sqrt{g_c(y)/g_c(x)}$ with $g_c(s) = s^3(c-s)$, since T_c is a primitive of $-c/(2\sqrt{g_c})$.

Lemma 46 (Gap system)**.** *A corona of $\{a_1, \dots, a_k\}$ in the disk R (all k circles wall-tangent) with cyclic order π exists iff the linear system*

$$\theta_{ij} \;\le\; S_j - S_i \;\le\; 2\pi - \theta_{ij} \qquad (1 \le i < j \le k,\; S_1 = 0)$$

is feasible, where θ_{ij} is the angle of the pair in positions i, j of π.

Proof. The angular separation of positions $i < j$ is $\min(S_j - S_i,\, 2\pi - (S_j - S_i))$, and $\min(\Delta, 2\pi - \Delta) \ge \theta \iff \theta \le \Delta \le 2\pi - \theta$ because $\theta \le \pi$; wall-tangent circles are contained in the disk and pairwise disjointness is exactly the angular condition. ◻

Lemma 47 (Subset certificates)**.** *If $\{a_1, \dots, a_k\}$ has a corona with cyclic order π, then every subset T, $|T| \ge 3$, satisfies $\sum_{\text{consecutive pairs of } T} \theta \le 2\pi$ in the induced cyclic order. Consequently, if some T exceeds 2π in* all *its cyclic orders, no corona exists in any order – the unconditional direction used by all lower bounds.* ◻

The naive criterion ("some order with consecutive sum $\le 2\pi$ implies a corona") is *false* for $k \ge 4$: the conditional triangle inequality $\theta(a,c) \le \theta(a,b) + \theta(b,c)$ holds when $b \ge \min(a,c)$ but fails badly for small b ($R = 1$, $a = c = 0.45$, $b \to 0$), and $\{0.47, 0.47, 0.47, 0.02\}$ has a small cyclic sum while its top trio is already infeasible. The correct statement:

Theorem 48 (Exact criterion per order, $k = 4$)**.** *For arbitrary symmetric $\theta_{ij} \in (0, \pi]$, the system of Lemma 46 for the order $(1,2,3,4)$ is feasible iff the four trio certificates and the total $\theta_{12} + \theta_{23} + \theta_{34} + \theta_{14} \le 2\pi$ hold.*

Proof. Necessity: Lemma 47. Sufficiency, by difference-constraint duality: each upper bound is an up-edge $i \to j$ of weight $2\pi - \theta_{ij}$, each lower bound a down-edge $j \to i$ of weight $-\theta_{ij}$; the system is feasible iff the digraph has no negative directed cycle, and it suffices to exclude *simple* cycles. A simple cycle Γ with U up-edges has weight $w(\Gamma) = 2\pi U - \sum_{e \in \Gamma} \theta_e$, so $w < 0$ requires $\#\text{edges}(\Gamma) > 2U$ (as $\theta \le \pi$). With $k = 4$, a simple cycle has ≤ 4 edges, so only $U = 1$ can go negative; a $U = 1$ cycle is an up-edge $i \to j$ plus a decreasing chain from j to i, and its negativity condition $\theta_{ij} + \sum_{\text{chain}} \theta > 2\pi$ is exactly the subset certificate of $\{i, \text{chain}, j\}$: enumerating, sizes 3 give the four trios and size 4 gives the total. ◻

Proposition 49 (The zigzag minimizes the total)**.** *For $a_1 \geq a_2 \geq a_3 \geq a_4$, the three cyclic orders have totals $\sum_{i<j} \theta_{ij}$ minus one perfect matching, and the maximum-sum matching is $\theta_{12}+\theta_{34}$; hence* $\min_\pi \mathrm{TOT}(\pi) = \sum_{i<j} \theta_{ij} - \theta_{12} - \theta_{34}$*, achieved by the zigzag order* (a_1, a_3, a_2, a_4)*.*

Proof. With $\sigma_i := \log f(a_i)$ (decreasing) one has $\theta(a_i, a_j) = g(\sigma_i + \sigma_j)$ for $g(s) = 2\arcsin(e^{s/2})$ on $s \leq 0$, with $g''(s) = e^{-s}/\big(2(e^{-s}-1)^{3/2}\big) > 0$: g is convex and increasing (continuity at $s = 0$ by passage to the limit). The three matchings have pair-sums $\{\sigma_1+\sigma_2, \sigma_3+\sigma_4\}$, $\{\sigma_1+\sigma_3, \sigma_2+\sigma_4\}$, $\{\sigma_1+\sigma_4, \sigma_2+\sigma_3\}$ with equal totals, and the first majorizes the other two; convexity turns majorization into the claimed maximality. □

Theorem 50 (Lemma U_4: global criterion, two inequalities)**.** $\{a_1 \geq a_2 \geq a_3 \geq a_4\}$ *admits a corona in the disk R (all pairs admissible) iff*

$$\theta_{12} + \theta_{13} + \theta_{23} \leq 2\pi \qquad \text{and} \qquad \sum_{i<j} \theta_{ij} - \theta_{12} - \theta_{34} \leq 2\pi.$$

Proof. Trio certificates do not depend on the order, so a corona exists iff all trios pass and the minimal total does (Theorem 48 + Proposition 49). By monotonicity of θ, every trio is dominated pairwise by the top trio $\{a_1, a_2, a_3\}$, so one trio inequality suffices. Neither inequality is redundant (both are active on the frontier). The top-trio condition linearizes by Lemma 45 with automatic hypothesis ($A = a_1$ is the global maximum): it reads $T_c(a_2) + T_c(a_3) \geq \tau_R$ with $c = R - a_1$. □

For $k = 5$ the subset-only criterion fails for exactly one pattern, the *pentagram*: of the 84 simple cycles, all with $U \geq 2$ are count-dominated or LP-dominated except the cycle through the five diagonals, and adding its certificate $\sum_D \theta \leq 4\pi$ restores an exact criterion for arbitrary θ; for *geometric* θ (rank-one matrices $\sin^2(\theta_{ij}/2) = f_i f_j$) the identity $\prod_D f_i f_j = \prod_{\text{sides}} f_i f_j = (\prod f_i)^2$ obstructs the pattern and the best value found is 7.556 against the $4\pi = 12.57$ needed; its redundancy for geometric θ is stated as a conjecture in the repository and is *not used*: lower bounds only ever use the unconditional direction of Lemma 47. For $k \geq 5$ the subset quantifier does not commute with the order quantifier (explicit five-ring counterexamples in the repository), so the per-order criterion or the LP is the correct tool.

D.2 Occupant walls and metallic floors

Lemma 51 (Walls, free-hole template)**.** *Suppose the holes of m, σ_1 and all o_i are free. If the exchange is blocked, then*

$$\textit{(B2)}\ \sigma_2 > 1 - \omega; \quad \textit{(B3)}\ \sigma_2 > \sigma_1 - \omega; \quad \textit{(B4)}\ \sigma_2 > \alpha - \omega - 1; \quad \textit{(Bo)}\ \sigma_2 > o_k - \omega\ \forall k;$$

$$\textit{(D)}\ \sigma_1 + \sigma_2 > 1; \qquad \textit{(W)}\ \sigma_1 + \sigma_2 \leq \alpha - \omega.$$

In particular $o_k < \sigma_2 + \omega \leq 1 + \omega$ for all k: every extra occupant stays within one width of m.

Proof. Each line is the contrapositive of an explicit placement whose two resources are disjoint and whose feasibility criteria (pair, nesting, row) are exact: $\sigma_2 \to H_m$ with $\sigma_1 \to D_m$; $\sigma_2 \subset \sigma_1 \to D_m$; σ_2 next to m in u ($1+\sigma_2 \leq \alpha-\omega$, pair-exact) with $\sigma_1 \to D_m$; $\sigma_2 \subset o_k$ with $\sigma_1 \to D_m$; the whole pair into D_m (a free unit disk). (W) is the legality of the witness's placement. □

Proposition 52 (Occupant price)**.** *Blocking in the free-hole template with $j \geq 1$ extra occupants implies*

$$\rho \;>\; 2+\frac{j-2\omega}{o_1} \;\geq\; \frac{j+2}{1+\omega} \quad (\omega \leq j/2), \qquad \textit{and for } j=1,\ \omega \geq \tfrac{1}{2}: \quad \rho > \frac{4}{1+2\omega}.$$

Proof. The tail of o_1 contains $\{o_2, \ldots, o_j, m, \sigma_1, \sigma_2\}$ (ties by the first-copy convention). By (Bo), $\sigma_1 \geq \sigma_2 > o_1 - \omega$, and each $o_i \geq 1$: $\rho \geq \big((j-1)+1+\sigma_1+\sigma_2\big)/o_1 > \big(j+2(o_1-\omega)\big)/o_1 = 2+(j-2\omega)/o_1$. If $\omega \leq j/2$ the expression decreases in o_1 and $o_1 < 1+\omega$ (Lemma 51). For $j = 1 < 2\omega$, add (D): $\sigma_1+\sigma_2 > \max(1, 2(o_1-\omega))$, so $\rho > \big(1+\max(1,2(o_1-\omega))\big)/o_1$, whose infimum over o_1 is $4/(1+2\omega)$ at $o_1 = \frac{1}{2}+\omega$. □

Corollary 53 (Crossings; canonical optimality)**.** *For $j = 1$: $\rho > 3/(1+\omega) > T$ for $\omega < 3/T - 1 = 3T^2 - 3T - 4 = 0.6310\ldots$, and combining with the fine branch, $\rho > T$ for all $\omega < 2/T - \frac{1}{2} = 2T^2 - 2T - \frac{5}{2} = 0.5873\ldots$, with $\rho \geq \frac{13}{7}$ for $\omega \leq \frac{15}{26}$. For $j \geq 2$: $\rho > 4/(1+\omega) > 2 > T$ for all $\omega < 1$. Moreover every extra-occupant blocking has $\rho > 2$ for $\omega < \frac{1}{2}$, while the entire canonical curve lies strictly below 2 – the witness branch by the exact identity $(2+\omega)^3 - (1+\omega)\big((2+\omega)^2+(2+\omega)+1\big) = 1 > 0$, i.e. $T_{1+\omega} < 2+\omega$, hence $\Phi(\omega) < 2$ – so adding occupants is strictly suboptimal for the adversary: the canonical template is the optimal v in this class.* □

D.3 Occupied holes: the opposite-disk lemma and the silver floor

Lemma 54 (Opposite disk)**.** *Let a disk have capacity c, let $\sigma \leq c$ be the largest piece and C a multiset of pieces $\leq \sigma$. If $\sum C \leq c - \sigma$, then $\{\sigma\} \cup C$ packs in the disk.*

Proof. Place σ wall-tangent with center $(-(c-\sigma), 0)$. The disk of radius $r = c - \sigma$ centered at $(c-r, 0)$ is wall-tangent and exactly tangent to σ (center distance $(c-\sigma)+(c-r) = c = \sigma + r$): a free disk of radius $c - \sigma$, in which C fits in a row (Lemma 3). □

Lemma 55 (General blocking wall Bo$''$)**.** *Blocking implies, for every node y: $y < \sigma_2 + \omega + X_y$.*

Proof. Otherwise $\sum$children$(y) \leq (y-\omega) - \sigma_2$ and Lemma 54 packs $\{\sigma_2\} \cup$ children(y) in the hole of y (if some child exceeds σ_2, apply the lemma with that child as the largest piece; the inequality is the same), with $\sigma_1 \to D_m$: unblocked. Positions are re-eligible, and inserting σ_2 (a ring $< m$) only changes the container of σ_2: legal. □

Theorem 56 (Silver floor; m without children)**.** *Blocking with $j \geq 1$, holes occupied arbitrarily at any depth, m childless, implies $\rho > \Psi(\omega) = (1-\omega) + \sqrt{(1-\omega)^2+1}$, the positive root of $u^2 - 2(1-\omega)u - 1$. In particular $\Psi(0) = 1+\sqrt{2}$ (the silver ratio), $\Psi(\frac{1}{4}) = 2$, and $\Psi > T \iff \omega < (T-1)^2/2 = 0.35220\ldots$*

Proof. Let y^* be a node of minimal radius; its children are all < 1 (a child ≥ 1 would be a smaller node), so $X := X_{y^*}$ is a sum of rings smaller than m, living in the tail of m. With $\sigma := \sigma_2 > 1-\omega$ (wall (B2), valid since H_m is free): the tail of m gives $\rho \geq \sigma_1+\sigma_2+X \geq 2\sigma + X$, and the tail of y^* with Lemma 55 gives $\rho \geq (1+\sigma_1+\sigma_2+X)/y^* > (1+2\sigma+X)/(\sigma+\omega+X)$. Minimize $F(\sigma, X) = \max\big(2\sigma+X,\ (1+2\sigma+X)/(\sigma+\omega+X)\big)$ over $\sigma \geq 1-\omega$, $X \geq 0$: for fixed σ the first term grows in X and the second decreases, so $F(\sigma,\cdot)$ is bounded below by the crossing value $u = 2\sigma + X$ with $u(u+\omega-\sigma) = 1+u$, i.e. the positive root of $u^2 - (1+\sigma-\omega)u - 1 = 0$ (when the crossing falls at $X < 0$, the same algebra shows $F(\sigma, 0) = 2\sigma$ already exceeds the root). The root increases in σ, so the program minimum is at $\sigma = 1-\omega$, giving $\Psi(\omega)$. □

Theorem 57 (Evacuation dichotomy; m with children)**.** *The conclusion of Theorem 56 holds without the hypothesis that m is childless.*

Proof. Evacuation to D_m: place σ_1 and all children of m in a row inside D_m (possible if $\sigma_1+M \leq 1$, $M := \sum \text{children}(m)$; children of σ_1 travel inside σ_1) and σ_2 in the emptied H_m (possible if $\sigma_2 \leq 1-\omega$). Contrapositive: blocking implies $\sigma_2 > 1-\omega$ *or* $\sigma_1 + M > 1$. In branch A the proof of Theorem 56 applies verbatim (it used only $\sigma_2 > 1-\omega$, Bo$''$ and the two tails; the mass $M \geq 0$ only fattens them). In branch B, let y^* be the minimal node, $s := \sigma_2 + X > 1-\omega$ (Bo$''$ and $y^* \geq 1$) and $A := \sigma_1+\sigma_2+M+X > 1+s$; the two tails give $\rho \geq A > 1+s$ and $\rho \geq (1+A)/y^* > (2+s)/(s+\omega)$, whose optimization yields the metallic mean $\Psi_B(\omega)$, the positive root of $u^2-(2-\omega)u-1$. Since the positive root of u^2-bu-1 increases in b and $2-\omega \geq 2(1-\omega)$, $\Psi_B \geq \Psi$: branch B is dominated. (The crossing $\Psi_B = T$ happens at $\omega = (T-1)^2$ – twice the branch-A threshold – via the *polynomial* identity $(T-1)^2T-(2T-T^2+1) = T^3-T^2-T-1$.) $\square$

Corollary 58. *In the canonical template ($j=0$) with m arbitrary: branch A gives the full curve $T_{\text{can}}(\omega)$ (its program uses (B2) only as an inequality), branch B gives $\rho \geq \Phi(\omega)$ (Proposition 41 uses only (B1) and (W)); in both, $\rho > \Phi(\omega) > T$ for all $\omega > 0$.* $\square$

D.4 The double-pocket wall, the golden line, and Ψ_j

Throughout this subsection $j \geq 1$ and the walls above are in force; write $X := X_{o_1}$, $M := \sum \text{children}(m)$, $X_\sigma := \sum \text{children}(\sigma_1)$, and add the tariffed-nesting wall (B3$'$): $\sigma_2 + X_\sigma > \sigma_1 - \omega$ (Lemma 54 applied to the hole of σ_1).

Lemma 59 (Double pocket)**.** *Blocking (template $j=1$, arbitrary occupancy) implies $\sigma_1 > b_2(\alpha, o_1)$, where $b_2(A,B) = AB(A+B)/(A^2+AB+B^2)$.*

Proof. Blocking implies $\{\alpha, o_1, \sigma_1, \sigma_2\}$ does not pack in the pan (full repacking is a legal resource: children travel inside their parents, positions are existential). Since $\alpha + o_1 \leq R$, non-packability is inherited by containment in the disk $\bar{R} = \alpha + o_1$, where the pair $\{\alpha, o_1\}$ is diametrically rigid: $|c_\alpha| \leq o_1$, $|c_{o_1}| \leq \alpha$, $|c_\alpha - c_{o_1}| \geq \alpha + o_1$ force equality throughout – the rescaled rigidity of Proposition 30. By the exact necessity of that proposition (rescaled), any third circle disjoint from both has radius $\leq b_2(\alpha, o_1)$, and there are *two* pockets, one on each side of the diameter, mutually disjoint with slack: their centers are at $(x_0, \pm y_0)$ with $y_0^2 - b_2^2 = 3b_2^2$, i.e. $y_0 = 2b_2$ exactly, so the mirror circles are $4b_2$ apart. If $\sigma_1 \leq b_2$ then also $\sigma_2 \leq b_2$, and placing each σ concentric in a pocket packs the quadruple in $\bar{R} \subseteq$ the pan: contradiction. $\square$

Theorem 60 (Golden line, both branches)**.** *Blocking in the template ($j=1$, arbitrary occupancy) implies $\rho > \varphi^2 - \frac{\varphi}{2}\omega$ for all ω in every case except the corner $\{$branch B, $\alpha < o_1$, $o_1 > \tilde{o} = 1.2956$, node-child, $\omega < \frac{1}{2}\}$, where the proved bound is $\geq 2 > T$. In particular $\rho > T$ for all $\omega < \omega_A = 2(\varphi^2 - T)(\varphi - 1) = 0.962585\ldots$*

Proof. Write $g := \sqrt{5}-1$. *Tail chain.* The tail of o_1 contains $\{m, \sigma_1, \sigma_2\} \cup \text{children}(o_1) \cup \text{children}(m) \cup \text{children}(\sigma_1)$, so $\rho\, o_1 \geq 1 + \sigma_1 + \sigma_2 + X + M + X_\sigma$; eliminating X with Bo$''$ ($X > o_1 - \sigma_2 - \omega$):

$$(*) \qquad \rho \;>\; 1 + \frac{1+\sigma_1-\omega+M+X_\sigma}{o_1}.$$

Branch A ($\sigma_2 \geq 1-\omega$). From (W), $\alpha \geq \sigma_1+\sigma_2+\omega \geq 1+\sigma_1$. Lemma 59 with b_2 increasing in α gives $\sigma_1 > b_2(1+\sigma_1, o_1)$, i.e. $o_1 < N(\sigma_1) := (1+\sigma_1)\big(\sqrt{1+4\sigma_1}-1\big)/2$ (the condition

$b_2(1+\sigma_1, N) = \sigma_1$ is the quadratic $N^2+(1+\sigma_1)N-\sigma_1(1+\sigma_1)^2 = 0$). With $(*)$ and $M, X_\sigma \geq 0$: $\rho > 1+(1+\sigma_1-\omega)/N(\sigma_1) =: 1+h(\sigma_1)$, and h is decreasing in σ_1: the comparison of derivatives reduces to the polynomial identity $(1+4\sigma_1)-(1+2\sigma_1-2\sigma_1^2)^2 = 4\sigma_1^3(2-\sigma_1) > 0$. The infimum is at $\sigma_1 = 1$: $N(1) = g$, $1/g = \varphi/2$, and $\rho > 1+(2-\omega)\varphi/2 = \varphi^2 - \frac{\varphi}{2}\omega$.

Branch B ($\sigma_1 + M > 1$). Using $\sigma_1 + M > 1$, Bo$''$, (B3$'$), (W) and Lemma 59:

$$\text{(I)}\quad \rho\, o_1 \geq 1+\sigma_1+\sigma_2+M+X+X_\sigma \;>\; 2+o_1-\omega+2b_2(\alpha, o_1)-\alpha,$$
$$\text{(II)}\quad \rho\,\alpha \geq o_1+1+\sigma_1+\sigma_2+M+X+X_\sigma \;>\; 2o_1+2-\omega+2b_2(\alpha, o_1)-\alpha \quad [\text{if } \alpha \geq o_1],$$

(in (I): $X > o_1-\omega-\sigma_2$, $X_\sigma > \sigma_1-\omega-\sigma_2$, $-\sigma_2 \geq \sigma_1+\omega-\alpha$ by (W), $\sigma_1 > b_2$; (II) is the tail of α, which contains o_1 and its children only when $\alpha \geq o_1$). Set $f_1 := 1+(2-\omega+2b_2-\alpha)/o_1$ and $f_2 := (2o_1+2-\omega+2b_2-\alpha)/\alpha$. If $o_1 \leq g$, the short chain (only $\sigma_1+M > 1$ and Bo$''$, valid in any order) gives $\rho > 1+(2-\omega)/o_1 \geq 1+(2-\omega)/g = \varphi^2 - \frac{\varphi}{2}\omega$. If $o_1 > g$, the admissible α-region is $[1+\omega, A_{\max}(o_1)]$ where $b_2(A_{\max}, o_1) = 1$, and two exact facts drive the optimization: b_2 is concave in α ($\partial^2 b_2/\partial\alpha^2 = -6\alpha o_1^3(\alpha+o_1)/D^3 < 0$, D the denominator), so f_1 is concave in α and minimized at the endpoints; and $D^2 - \alpha o_1^2(o_1+2\alpha) = (\alpha+o_1)(\alpha^3+\alpha^2 o_1+o_1^3) > 0$ gives $\alpha\,\partial b_2/\partial\alpha < o_1$, so f_2 is strictly decreasing in α. For $o_1 \in (g, \tilde{o}]$ ($\tilde{o} = 1.29558\ldots$; any order of α, o_1, only (I) used): at $\alpha = A_{\max}$, $f_1 - \text{curve} = c_{10}(o_1) + \omega(1/g - 1/o_1)$ with the ω-coefficient ≥ 0 and $c_{10} \geq 0$ on $[g, o^* = 1.5958]$, with the exact identity $f_1 \equiv$ curve at $o_1 = g$ (the golden corner, for all ω); at $\alpha = 1+\omega$, $f_1 \geq$ curve on $[g, o^*] \times [0,1]$ with contact only at $(g, \omega \to 1)$. For $o_1 > \tilde{o}$ with $\alpha \geq o_1$ (forcing $o_1 \leq \frac{3}{2}$, since $A_{\max}$ is decreasing with $A_{\max}(\frac{3}{2}) = \frac{3}{2}$ exactly; also $A_{\max}(g) = 2$, $A_{\max}(2) = g$ – the corner is self-dual): $f_2 \geq f_2(A_{\max}) = c_{20}(o_1) + \omega\, c_{21}(o_1)$ with $c_{21} \geq 0$ for $o_1 \leq 2$ and $c_{20} \geq 0$ on $[\tilde{o}, \frac{3}{2}]$. For $o_1 > \tilde{o}$ with $\alpha < o_1$ (a region requiring $\omega < \frac{1}{2}$: it needs $1+\omega \leq \alpha < o_1 < N_1(\omega)$, and $N_1(\frac{1}{2}) = \frac{3}{2}$ exactly with N_1 decreasing, where $b_2(1+\omega, N_1) = 1$): if the children of o_1 are all < 1, the two α-free bounds $\rho > 1+o_1-\omega$ (tail of m: $\sigma_1+M > 1$ and Bo$''$) and $\rho > 1+\big(2-\omega+2b_2(1+\omega, o_1)\big)/o_1$ (tail of o_1, which now contains α; Bo$''$, (B3$'$), (W) cancel α, and Lemma 59 applies with $\alpha \geq 1+\omega$) have max $\geq$ curve $+\,0.307$ throughout the region (certified on a grid with symbolic endpoints); if o_1 has a node child we are in branch B with $\omega < \frac{1}{2}$, and Theorem 57 gives $\rho > \Psi_B(\omega) > \Psi_B(\frac{1}{2}) = 2 > T$ ($\Psi_B(\frac{1}{2}) = 2$ exact). $\square$

Proposition 61 (The Ψ_j ladder, unconditional)**.** *Blocking with $j \geq 1$ extra occupants and arbitrary nested occupancy implies*

$$\rho \;>\; \Psi_j(\omega) = (1-\omega) + \sqrt{(1-\omega)^2 + j} \qquad (\textit{root of } u^2 - 2(1-\omega)u - j),$$

with $\Psi_j > T \iff \omega < 1-(T^2-j)/(2T)$: 0.3522 ($j=1$), 0.6240 ($j=2$), 0.8959 ($j=3$), and all ω for $j \geq 4$ ($\Psi_j \geq \sqrt{j} \geq 2$).

Proof (leaf argument). Call a node a *leaf* if it has no node children. The node subtree of each occupant o_i is finite and non-empty, hence contains a leaf ℓ_i; the subtrees are disjoint, so there are j distinct leaves $\ell_1, \ldots, \ell_j \geq 1$. Let L be the largest, $X_L = \sum \text{children}(L)$ (all < 1, as L is a leaf), $M = \sum \text{children}(m)$, and W the total mass of rings < 1 other than σ_1, σ_2 (so $X_L \leq W$, $M \leq W$, and X_L, M are disjoint). Three facts: (1) *wall at the leaf* (Lemma 55): $L < \sigma_2+\omega+X_L \leq \sigma_2+\omega+W$; (2) *tail of L*: the other $j-1$ leaves, m, σ_1, σ_2 and all mass < 1 are $\leq L$ (first-copy convention), so $\rho L \geq (j-1)+1+\sigma_1+\sigma_2+W \geq j+2\sigma_2+W$; (3) *tail of m*: $\rho \geq \sigma_1+\sigma_2+W \geq 2\sigma_2+W$. In branch A of the evacuation dichotomy ($\sigma_2 \geq 1-\omega$), facts (1)–(3) give $\rho > \max\big(2\sigma+W,\ (j+2\sigma+W)/(\sigma+\omega+W)\big)$ at $\sigma = \sigma_2$; minimizing over $\sigma \geq 1-\omega$,

$W \geq 0$, the crossing $u^2 + u(\omega - \sigma - 1) - j = 0$ ($u = 2\sigma + W$) increases in σ, with minimum at $\sigma = 1 - \omega$: exactly $\Psi_j(\omega)$. (If the crossing falls at $W < 0$ – possible for $\omega > \frac{1}{2}$, $j = 1$, $\sigma \to 1$ – the value 2σ at $W = 0$ satisfies $(2\sigma)^2 - 2(1-\omega)(2\sigma) - j \geq j > 0$, so $2\sigma > \Psi_j$ anyway.) In branch B ($\sigma_1 + M > 1$), set $s := \sigma_2 + X_L$: by fact (1) and $L \geq 1$, $s > 1 - \omega$; the tail of L also contains M, disjoint from X_L, so $\rho L \geq j + \sigma_1 + M + \sigma_2 + X_L > j + 1 + s$ with $L < s + \omega$, and the tail of m gives $\rho \geq \sigma_1 + M + \sigma_2 + X_L > 1 + s$; minimizing $\max\bigl(1 + s,\ (j + 1 + s)/(s + \omega)\bigr)$ yields, with $u = 1 + s$, the metallic mean $u^2 - (2 - \omega)u - j = 0$, whose root dominates Ψ_j ($2 - \omega \geq 2(1 - \omega)$, and the root increases in the linear coefficient). ☐

The leaf argument is what closes the case caught by the adversarial verification (a node child inside the largest occupant): towers of nested nodes inflate X_{o_1} without paying the tail of m, but they always leave a leaf at the bottom of each subtree, and the largest leaf carries both the j-bonus in its tail and the leaf ceiling $\sigma_2 + \omega + W$.

Corollary 62 (Small rings are free for the combinatorial walls)**.** *Proposition 52, Theorems 56 and 57 and Proposition 61 hold unchanged if the instance contains additional rings smaller than m anywhere compatible with each theorem's template (with S still the pair $\{\sigma_1, \sigma_2\}$).*

Proof. Every unblocking placement used by those walls is *local*: it lives in D_m (freed by the exchange), in H_m, in the holes of nodes or of σ_1, or next to m inside u – never in the free space of v. An extra ring smaller than m, wherever it sits, intersects none of these resources (hole contents are already counted in the X's; rings in v proper stay put, which is legal since no other position changes), and tails can only grow. (The geometric wall of Lemma 59 is *not* covered: it uses the global repacking of the pan, which does see small rings in v; this is the "gap lemma" left open.) ☐

D.5 Three-piece profiles in the canonical template

Here v is the canonical pan of radius $R = \alpha + 1$ with $\{\alpha, m\}$ diametral, u is the hole of α, and the witness placed $S = \{\sigma_1 \geq \sigma_2 \geq \sigma_3\} \subset (\omega, 1)$ in u; the hole of m may carry mass $M \geq 0$. The reinsertion resources are D_m (rows of sum ≤ 1, with α in place), H_m (capacity $1 - \omega$; its content M can be evacuated in a row to D_m), the slot next to m in u (row $1 + \Sigma \leq \alpha - \omega$), nesting inside S, and the repack of v: placing $\{\alpha\} \cup$ (roots of S) as a corona in the pan – incompatible with using D_m or the evacuation, both of which pin α. The witness gives (W) $\sigma_1 + \sigma_2 \leq \alpha - \omega$.

Theorem 63 (Three pieces; the zigzag pays)**.** *Blocking with S a triple and arbitrary M implies $\rho > \max\bigl(\Phi(\omega), \frac{13}{7}\bigr)$ for all $\omega \in (0, 1)$; in the zigzag branch below, $\rho > \frac{17}{7}$.*

Proof. Branch 1 ($\sigma_3 \leq \sigma_1 - \omega$): the dust rides the pair. Nest σ_3 in σ_1 inside each pair placement; their failures give five walls: (B1) "$\sigma_3 \subset \sigma_1$; corona $\{\alpha, \sigma_1, \sigma_2\}$ in v" fails, so the trio has no corona and $F(\sigma_1, \sigma_2) > 2\pi$ (corona $\iff$ $F \leq 2\pi$ is exact for three circles); (B2/dichotomy) "$\sigma_1(\sigma_3) + M$ in a row in D_m; $\sigma_2 \to$ emptied H_m" fails, so $\sigma_2 > 1 - \omega$ or $\sigma_1 + M > 1$; (B4) "$\sigma_2 \to u$ next to m; corona $\{\alpha, \sigma_1(\sigma_3)\}$" fails, so $\sigma_2 > \alpha - \omega - 1$; (BH) "$\sigma_1(\sigma_3) \to D_m$; $\sigma_2 + M$ in a row in H_m" fails, so $\sigma_2 + M > 1 - \omega$; and (W). From (B1)+(W), by Corollary 40, in *both* branches of the dichotomy $\sigma_1 + \sigma_2 \geq 1 + b(\alpha)$, $\alpha \geq T_{1+\omega}$ and $\rho \geq \sigma_1 + \sigma_2 \geq \Phi(\omega) > T$. Moreover: if $\sigma_2 > 1 - \omega$, the program (B1)+(B2)+(B4)+(W) is exactly that of Theorem 21, whose lower bound gives $\sigma_1 + \sigma_2 > \frac{13}{7}$; if instead $\sigma_1 + M > 1$ and $\sigma_3 > \omega$, then (BH) gives $\rho \geq \sigma_1 + \sigma_2 + \sigma_3 + M > \sigma_1 + (1 - \omega) + \omega = 1 + \sigma_1 \geq 1 + \Phi(\omega)/2 \geq 1 + T/2 = 1.9196 > \frac{13}{7}$.

Branch 2 ($\sigma_3 > \sigma_1 - \omega$): nothing nests (also $\sigma_2, \sigma_3 > \sigma_2 - \omega$ and the star needs $\sigma_2 + \sigma_3 \le \sigma_1 - \omega$, impossible). Split by the trio.

2A: the trio $\{\alpha, \sigma_1, \sigma_2\}$ has no corona. Then $F > 2\pi$ and, as above, $\rho \ge \sum S > \sigma_1 + \sigma_2 \ge \Phi(\omega)$. Furthermore $\sigma_1 \ge (\sigma_1 + \sigma_2)/2 \ge \Phi/2$, so $\sigma_3 > \sigma_1 - \omega \ge \Phi/2 - \omega$ and $\rho > \frac{3}{2}\Phi - \omega$ when $\omega \le \Phi/2$; by the chord bound $\Phi \ge T + (13 - 7T)\omega$ this exceeds 2.64 on $(0, \frac{1}{7}]$, while for $\omega \ge \frac{1}{7}$, $\Phi \ge \frac{13}{7}$: in all cases $\rho \ge \frac{13}{7}$.

2B: the trio has a corona. Three mixed placements fail: "$\sigma_3(+M)$ in a row in H_m + trio corona" gives (m1) $\sigma_3 + M > 1 - \omega$; "$\sigma_3 \to u$ next to m + trio corona" gives (m2) $\alpha < 1 + \omega + \sigma_3$; and "corona of the quadruple" fails, so by Theorem 50 – the top trio passing its certificate, by Lemma 47 – the *zigzag* fails: $\theta(\alpha, \sigma_2) + \theta(\alpha, \sigma_3) + \theta(\sigma_1, \sigma_2) + \theta(\sigma_1, \sigma_3) > 2\pi$. By monotonicity, $2[\theta(\alpha, \sigma_2) + \theta(\sigma_1, \sigma_2)] > 2\pi$, so $A + B > \pi/2$ with $A = \theta(\alpha, \sigma_2)/2$, $B = \theta(\sigma_1, \sigma_2)/2$, both $< \pi/2$ ($A \ge \pi/2$ would force $\sigma_2 \ge 1$; $\sigma_1 + \sigma_2 < R$). Hence $\sin A > \cos B$, i.e. $\sin^2 A + \sin^2 B > 1$: $f(\sigma_2)\big[f(\alpha) + f(\sigma_1)\big] > 1$ with $f(\alpha) = \alpha$ and $f(\sigma_1) \le 1/\alpha$, so $f(\sigma_2)(\alpha + 1/\alpha) > 1$, which is *exactly* $\sigma_2 > b(\alpha)$. With (W): $2b(\alpha) < \sigma_1 + \sigma_2 \le \alpha - \omega$, and since $\alpha \mapsto \alpha - \omega - 2b(\alpha)$ is strictly increasing ($1 - 2b' > 0$ for $\alpha \ge 1$), $\alpha > \gamma_\omega$, the root of $2b(\alpha) = \alpha - \omega$ – equivalently of $\alpha^3 = \alpha^2 + \alpha + \omega(\alpha^2 + \alpha + 1)$, with $\gamma_0 = \varphi$. The tails: by (m1), $\rho \ge \sum S + M > 2b(\alpha) + (1 - \omega) \ge \gamma_\omega + 1 - 2\omega$, and by (m2), $\rho > 2b(\alpha) + \alpha - 1 - \omega \ge 2\gamma_\omega - 1 - 2\omega$. The first bound decreases in ω and the second increases ($1 < \gamma' < 2$), and they cross *exactly* at $\gamma = 2$, $\omega = \frac{2}{7}$ ($8 = 4 + 2 + 2$ in the cubic), where both equal $3 - \frac{4}{7} = \frac{17}{7}$: $\rho > \frac{17}{7}$ throughout branch 2B. □

Proposition 64 (Reduction branch with occupants)**.** *In the templates with $j \ge 1$ extra occupants and S of $k \ge 3$ pieces with $\sum_{i \ge 3} \sigma_i \le \sigma_1 - \omega$, all pair walls and bounds (Proposition 52, Theorems 56 and 57, Proposition 61) hold unchanged: blocking implies $\rho > \Psi_j(\omega)$.*

Proof. Nest $\sigma_3, \ldots, \sigma_k$ in a row inside the hole of σ_1; every unblocking placement of those walls moves σ_1 (with its cargo travelling inside) and σ_2 exactly as in the pair case, and the tails only grow. □

D.6 The golden floor of the pan exchange

Here u is the pan itself: F places m at top level, the witness P keeps m in the hole of a ring y ($1 + X_y^{\text{rest}} \le y - \omega$), the pan holds occupants $O = \{o_1 \ge \cdots \ge o_j\}$ (> 1, top level, shared by F and P, arbitrarily occupied) and, per P, the pair $S = \{\sigma_1 \ge \sigma_2\}$. Resources: D_m (free unit disk in y's hole), H_m, hole tariffs (Lemma 54), nesting, and the pan repack. F's pan gives the pair necessities $o_i + 1 \le R$ and $o_i + o_k \le R$. *Strict leaf*: a leaf whose hole is not y's; with $j \ge 2$ at least $j - 1$ exist (j if y is not a leaf).

Theorem 65 (Golden floor, pan exchange, pair profiles)**.** *Blocking implies $\rho > \varphi$ – for $j = 1$ and $j = 2$ at every width and occupancy, for the evacuation branch ($\sigma_1 + M > 1$) at every width, and for $j \ge 3$ through the strict-leaf ladder and a case tree, except one declared micro-cell for $j = 3$ (solid-disk $\sigma_2 \le \omega$, $\omega \ge \varphi/2$, y a leaf, $o_1 \ge 3$; directed evidence* $\min \rho = 2.37$*). The infimum is exactly φ, realized by the golden family of Theorem 19 inside the case $j = 1$.*

Proof. $j = 1$. The placement "$\sigma_i \to$ pan, the other to D_m" fails, so $\{o_1, m, \sigma_i\}$ does not pack in R, hence (antitone containment, $R \ge o_1 + 1$) not in the disk $o_1 + 1$, where $\{o_1, 1\}$ is rigid: $\sigma_i > b(o_1)$ by Proposition 30. The tails give $\rho > \max\big(2b(o_1), (1 + 2b(o_1))/o_1\big)$; b is increasing, $g(A) = (1 + 2b)/A$ decreasing (the certificate $(3A^2 + 3A + 1)(A^2 + A + 1) - 2A(2A + 1)$ has positive coefficients), and the crossing $2b(A)A = 1 + 2b(A)$ factors as $(A^2 - A - 1)(2A + 1)$: unique positive root $A = \varphi$, value φ ($2b(\varphi) = \varphi$, $g(\varphi) = (1 + \varphi)/\varphi = \varphi$).

$j=2$*: the mirror-pocket wall.* The same placement fails, so $\{o_1, o_2, m, \sigma_2\}$ does not pack in R, hence not in the disk o_1+o_2, where $\{o_1, o_2\}$ is rigid and the quadruple packs *iff* $m \le b_2(o_1, o_2)$ and $\sigma_2 \le b_2$ (necessity by Proposition 30 per circle; sufficiency by placing them concentric in the two mirror pockets, disjoint since $y_0 = 2b_2$, as in Lemma 59). Thus $b_2(o_1, o_2) < 1$, i.e. $o_2 < \bar{A}(o_1)$, the decreasing root of $b_2(o_1, \cdot) = 1$ with $\bar{A}(2) = \sqrt{5} - 1$ and $\bar{A}(\frac{3}{2}) = \frac{3}{2}$. The tails of o_1 and o_2, with (D) $\sigma_1 + \sigma_2 > 1$ (row in D_m), give $\rho > \max\big((o_2+2)/o_1,\ 2/o_2\big)$ on $o_2 \in (1, \bar{A}(o_1))$: for $o_1 \le \frac{3}{2}$, $3/o_1 \ge 2$; for $o_1 \in (\frac{3}{2}, 2)$ the crossing $o_2^* = \sqrt{1+2o_1} - 1$ is interior ($b_2(o_1, o_2^*) < 1$ there, with exact equality at $o_1 = 2$) and its value $2/o_2^* > 2/(\sqrt{5}-1) = \varphi$; for $o_1 \ge 2$, $2/o_2 > 2/\bar{A}(2) = \varphi$. The crossing at $o_1 = 2$ is the golden corner $b_2(2, \sqrt{5}-1) = 1$ once more.

Evacuation branch, $j \ge 2$. A strict leaf L' with $s := \sigma_2 + X_{L'} > 1 - \omega$ satisfies (Lemma 54) $L' < s + \omega$, and its tail collects $\{m, \sigma_1, \sigma_2, M\} \cup \mathrm{children}(L')$: with $\sigma_1 + M > 1$, exactly the branch-B program of Theorem 57, so $\rho > \Psi_B(\omega)$, and $\Psi_B(1) = \varphi$ *exactly* with Ψ_B decreasing: $\rho > \varphi$ for all $\omega < 1$.

$j \ge 3$*, remaining branch.* The strict-leaf ladder gives $\rho > \Psi_{jj}(\omega)$ with $jj \in \{j-1, j\}$ leaves, and the golden crossings are exact: $\Psi_1(\frac{1}{2}) = \varphi$, $\Psi_2(\varphi/2) = \varphi$, $\Psi_3(1) = \sqrt{3} > \varphi$ – covering $j \ge 4$ outright and $j = 3$ up to $\varphi/2$ (always if y is not a leaf). The rest of $j = 3$ is a case tree: $o_2 < 3/\varphi$ (tail of o_2 exceeds $3/o_2$), or $o_1 < 3$ (tail of o_1 exceeds $(3/\varphi+3)/o_1$, and $(3/\varphi+3)/\varphi = 3$ exactly), or $o_1 \ge 3$ with the node/dust dichotomy on o_1's hole content: a node makes y's subtree or o_1's contribute a third strict leaf (Ψ_3), dust lands in m's tail ($\rho \ge \sigma_1 + \sigma_2 + X_1 > o_1 - \omega \ge 2$ when $y = o_1$, and $> o_1 - 1$ otherwise); the declared micro-cell is what survives this dichotomy when $\sigma_2 \le \omega$ (a solid disk) disables the $\rho > 2\omega$ shortcut. Full details, the adversarial reports (two rounds, including two repaired steps), and the tower-argument sketch for the micro-cell are in the repository (`code/batalla2.py`, 6/6). $\square$

E Verification details for Theorem 14

All feasibilities in the twin instances are two-circle-exact except three facts, proved here; every numerical claim is also reproduced by the scripts in the repository (`code/gemelas.py`).

(V1) $\{10, 4.99, 4.50\}$ **does not pack in a disk of radius 15.** As in Lemma 12: $|c_A| \le 5$, $|c_X| \le 10.01$, $|c_Y| \le 10.5$, $|c_A - c_X| \ge 14.99$, $|c_A - c_Y| \ge 14.5$, $|c_X - c_Y| \ge 9.49$; the first two give $a := |c_A| \ge 4.98$. With $u = -c_A/|c_A|$, $c_X \cdot u \ge (124.5 - a^2)/(2a) \ge 9.95$ and $c_Y \cdot u \ge (100 - a^2)/(2a) \ge 7.5$ (both decreasing in a, evaluated at $a = 5$), with transverse norms at most $\sqrt{10.01^2 - 9.95^2} \le 1.095$ and $\sqrt{10.5^2 - 7.5^2} \le 7.349$. Hence $|c_X - c_Y|^2 \le 10.01^2 + 10.5^2 - 2(9.95 \cdot 7.5 - 1.095 \cdot 7.349) \le 77.30$, so $|c_X - c_Y| \le 8.80 < 9.49$.

(V2) With the pan pair $\{10, 5\}$ **placed, no third circle of radius** $z \in \{4.74, 4.76\}$ **fits.** From $|c_A| \le 5$, $|c_B| \le 10$ and $|c_A - c_B| \ge 15$ the triangle inequality forces equalities: the pair is diametrically rigid, $c_B = 10u$ for a unit vector u and $c_A = -5u$. For a third circle of radius z: $c_z \cdot u \ge \big((10+z)^2 - (15-z)^2 - 25\big)/10$, which equals 8.8 for $z = 4.76$ and 8.7 for $z = 4.74$; then $|c_z - c_B|^2 = |c_z|^2 - 20\, c_z \cdot u + 100 \le (15-z)^2 - 20\, c_z \cdot u + 100$, giving $|c_z - c_B| \le 5.38 < 9.76$ and $\le 5.60 < 9.74$ respectively.

(V3) $\{10,\, 4.76,\, 4.74\}$ **packs in a disk of radius 15.** Explicit boundary configuration: $c_A = (-5, 0)$,

$$c_X = 10.24(\cos 30.77°,\ \sin 30.77°) \approx (8.799,\ 5.238),$$
$$c_Y = 10.26(\cos 32.02°,\ -\sin 32.02°) \approx (8.700,\ -5.440).$$

Then $|c_A| = 5$, $|c_X| = 10.24 = 15 - 4.76$, $|c_Y| = 10.26 = 15 - 4.74$ (all inside); $|c_A - c_X| = 14.76$ and $|c_A - c_Y| = 14.74$ (exact tangencies, chosen via the boundary angles); and $|c_X - c_Y| \approx 10.68 \geq 9.50$ with ample margin.

(V4) Hole facts. The hole of the 10 has radius 9.495; $4.99 + 4.50 = 9.49 \leq 9.495$ (the pair fits, two-circle exact) while $4.76 + 4.74 = 9.50 > 9.495$ and $5 + 4.50 = 9.50 > 9.495$ (blocked, two-circle exact).

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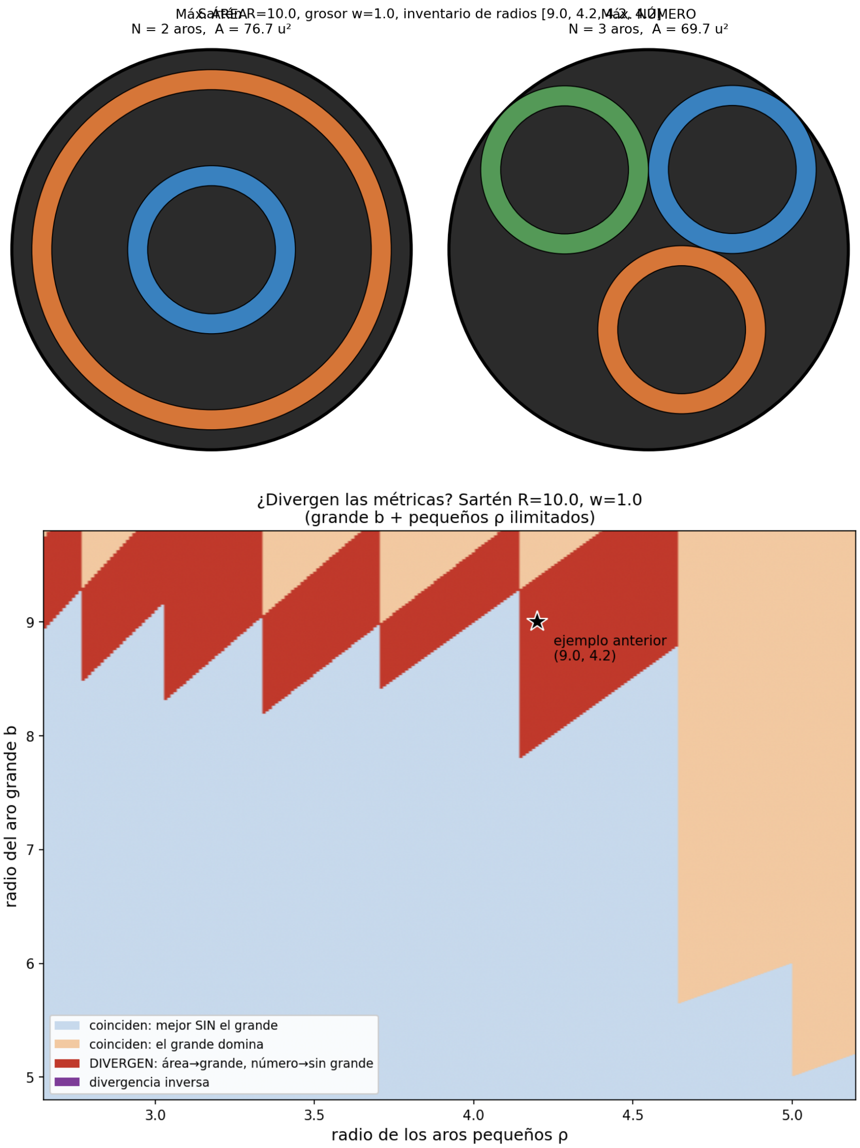

Figure 1: Top: the minimal divergence instance; area optimum (left) versus cardinality optimum (right). Bottom: phase diagram of the divergence band for uniform width.

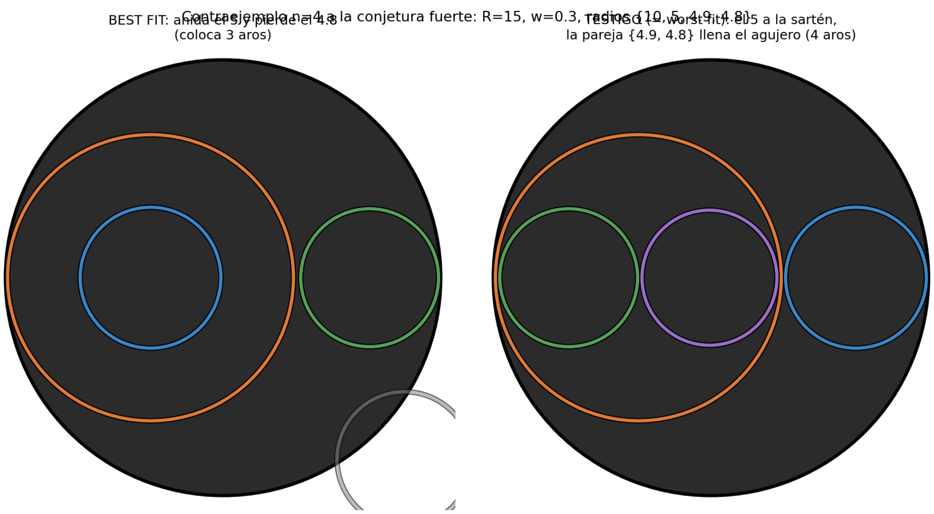

Figure 2: The $n = 4$ counterexample of Theorem 13: best fit (left) nests the 5 and loses the 4.8; the witness (right), realized by worst fit, places all four rings.